

Module 20

Cryptography

Cryptography

Cryptography is the science of **securing data** through **mathematical techniques**, so that only the intended recipient can read or use the information.

Purpose of Cryptography

Function	Description
 Confidentiality	Ensures only authorized parties can read the data
 Integrity	Ensures the data hasn't been altered or tampered
 Authentication	Verifies the identity of the sender
 Non-repudiation	Prevents denial of actions (e.g., "I didn't send that")

Types of Cryptography

1. Symmetric Key Cryptography

- **Same key** is used to **encrypt and decrypt** the data.
- Fast and efficient but key sharing is a problem.

Example Algorithms:

- AES (Advanced Encryption Standard)
- DES (Data Encryption Standard)

Example:

Key: 1234

Plain: HELLO → Cipher: 89GH (Encrypted with 1234)

2. Asymmetric Key Cryptography (Public-Key)

- Uses a **pair of keys**: Public Key (for encryption), Private Key (for decryption).
- Solves the problem of key sharing.

Example Algorithms:

- RSA
- ECC (Elliptic Curve Cryptography)

Example:

Public Key → Encrypt → 

Private Key → Decrypt → 

3. Hashing

- Converts data to a fixed-length "hash" (digest).
- **One-way function**: cannot be reversed.
- Used for passwords, data integrity, and digital signatures.

Example Algorithms:

- MD5 (not secure anymore)
- SHA-256 (commonly used)

Example:

"password123" → SHA-256 →
"ef92b778bafef771e89245b89ecbc3..."

Common Use Cases

Use Case	Example
 Secure communication	Emails encrypted with PGP/GPG
 Password storage	Stored as hashes (not plain text)
 Online transactions	SSL/TLS encryption in browsers
 Digital signatures	Verify document authenticity
 File encryption	Encrypt files with tools like GPG

Common Attacks on Cryptography

Attack Type	Description
 Brute force	Trying all possible keys
 Dictionary attack	Using common words as keys or passwords
 Man-in-the-middle	Intercepting data between two parties
 Replay attack	Resending captured messages or credentials

Summary

- Cryptography protects data by transforming it using keys and algorithms.
 - It enables **secure communication, privacy, authentication, and integrity**.
 - Widely used in **cybersecurity, ethical hacking, cloud, and web applications**.
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There are two types of cryptography

1. Symmetric Key Cryptography

- **Same key** is used for both **encryption and decryption**.
- Fast and efficient for large data.
- **Key must be shared securely** between sender and receiver.

Example:

AES (Advanced Encryption Standard)

 **Use case:** Encrypting files or communication between trusted systems.

2. **Asymmetric Key Cryptography (Public-Key)**

- Uses a **pair of keys**:
 - **Public key** to encrypt
 - **Private key** to decrypt
- More secure for data exchange, but slower.

Example:

RSA, ECC

 **Use case:** Secure email, SSL/TLS (HTTPS), digital signatures.

❖ **Perform encryption using cryptoforge**

What is CryptoForge?

CryptoForge is a **data encryption software** that helps you **secure your files and sensitive information** using strong cryptographic algorithms like:

- **AES (Advanced Encryption Standard)**
- **Blowfish**
- **Triple DES**
- **Gost**

It's mainly used to **encrypt/decrypt files, folders, and text**, keeping them **safe from unauthorized access**.

Why Use CryptoForge?

It protects your data by converting it into unreadable code unless the correct password (called a **passphrase**) is used. This is helpful when:

- You store sensitive documents on your PC or cloud.
 - You want to send confidential files over email.
 - You need to securely delete files.
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Key Features

Feature	Description
File Encryption	Encrypt files/folders using right-click or the app.
Text Encryption Tool	Encrypt/decrypt messages and clipboard content.
Shredder	Securely delete files (no one can recover them).
Strong Algorithms	Choose from AES, Blowfish, 3DES, etc., up to 448-bit keys.
Multiple File Support	Encrypt multiple files at once.
Passphrase-Based	Uses passwords instead of key files for easy handling.

How It Works (In Simple Steps)

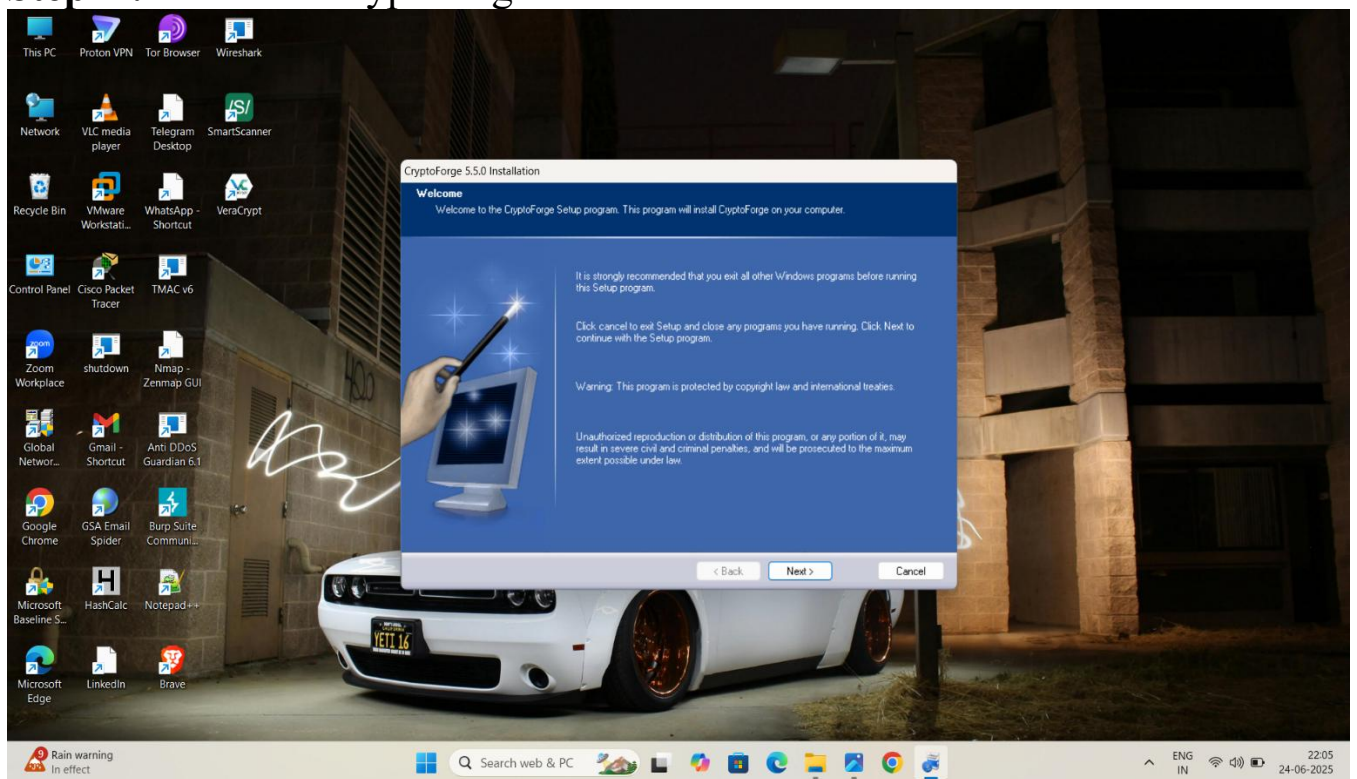
1. **Choose a file** → Right-click → **Encrypt with CryptoForge**
 2. **Set a passphrase** → CryptoForge encrypts the file
 3. To access the file again → Right-click → **Decrypt** → Enter passphrase
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Where It's Used

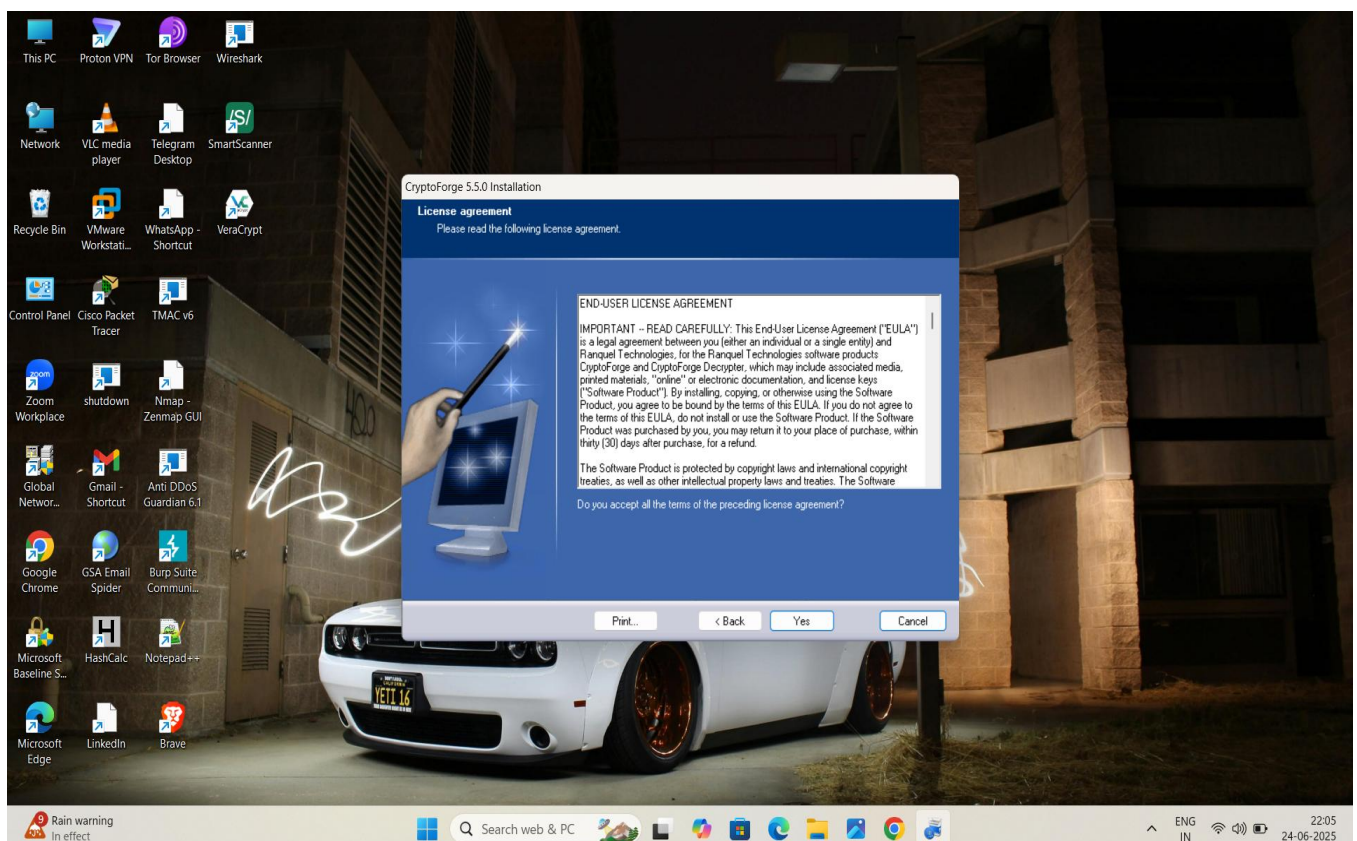
- Personal security (photos, documents)
 - Businesses protecting client data
 - Sending secure emails or USB data
 - Avoiding ransomware risks by securing backups
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How to use :

Step 1 : install the cryptoforge

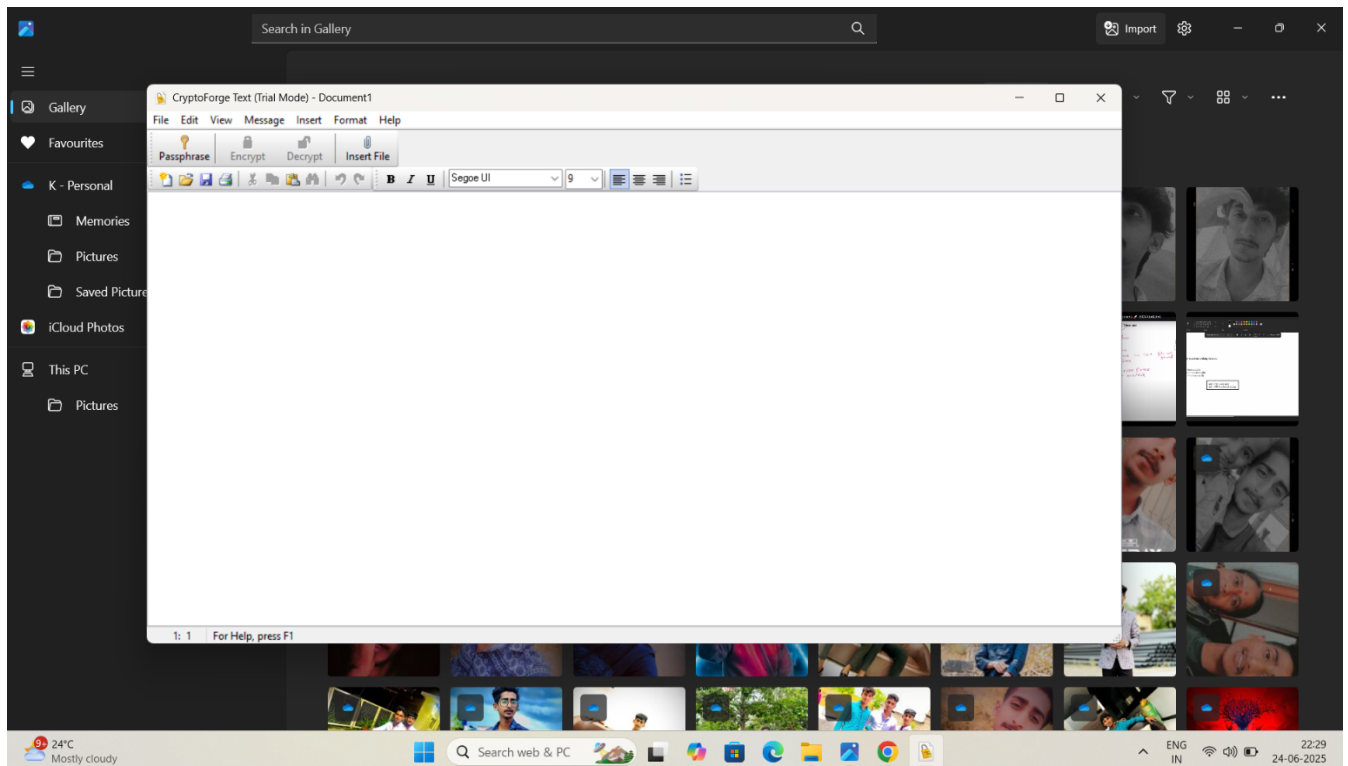


2nd step : click on next

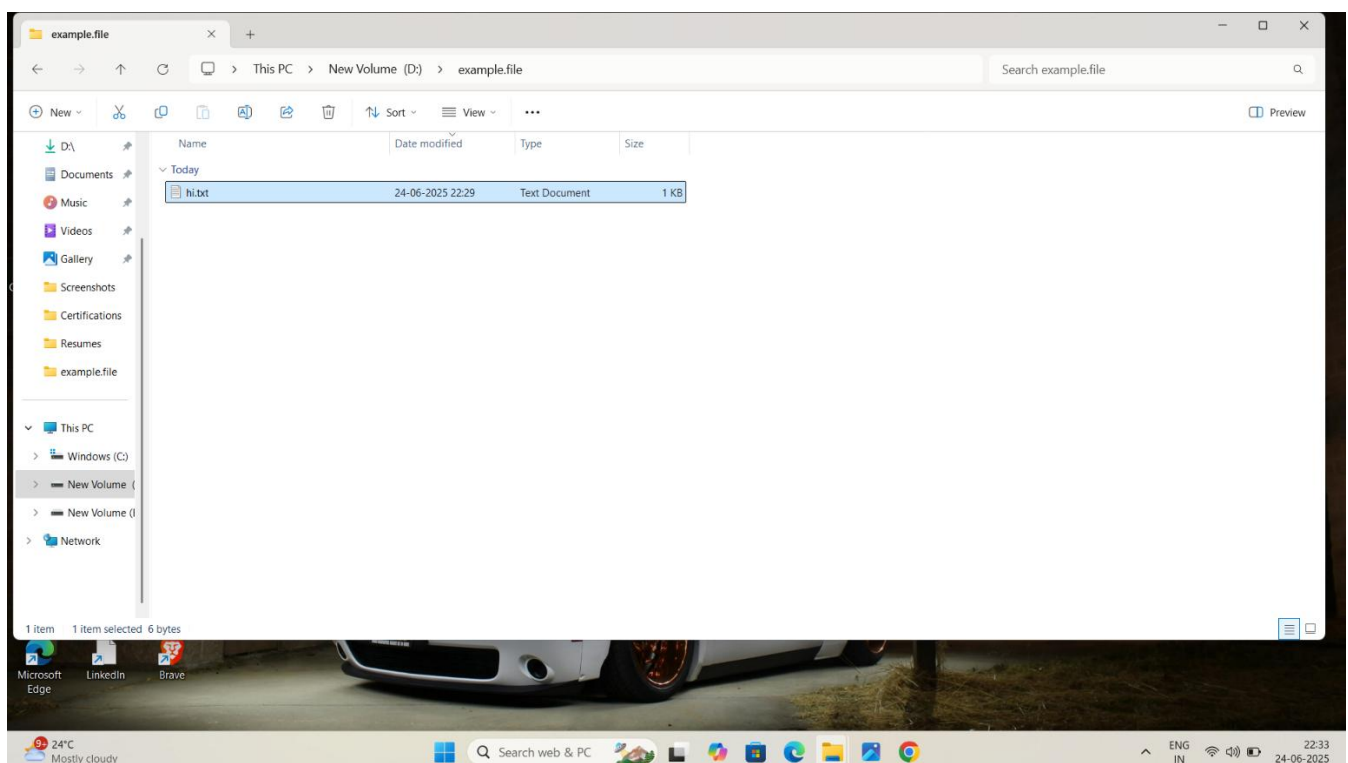


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3rd step : you can see the cryptoforge interface

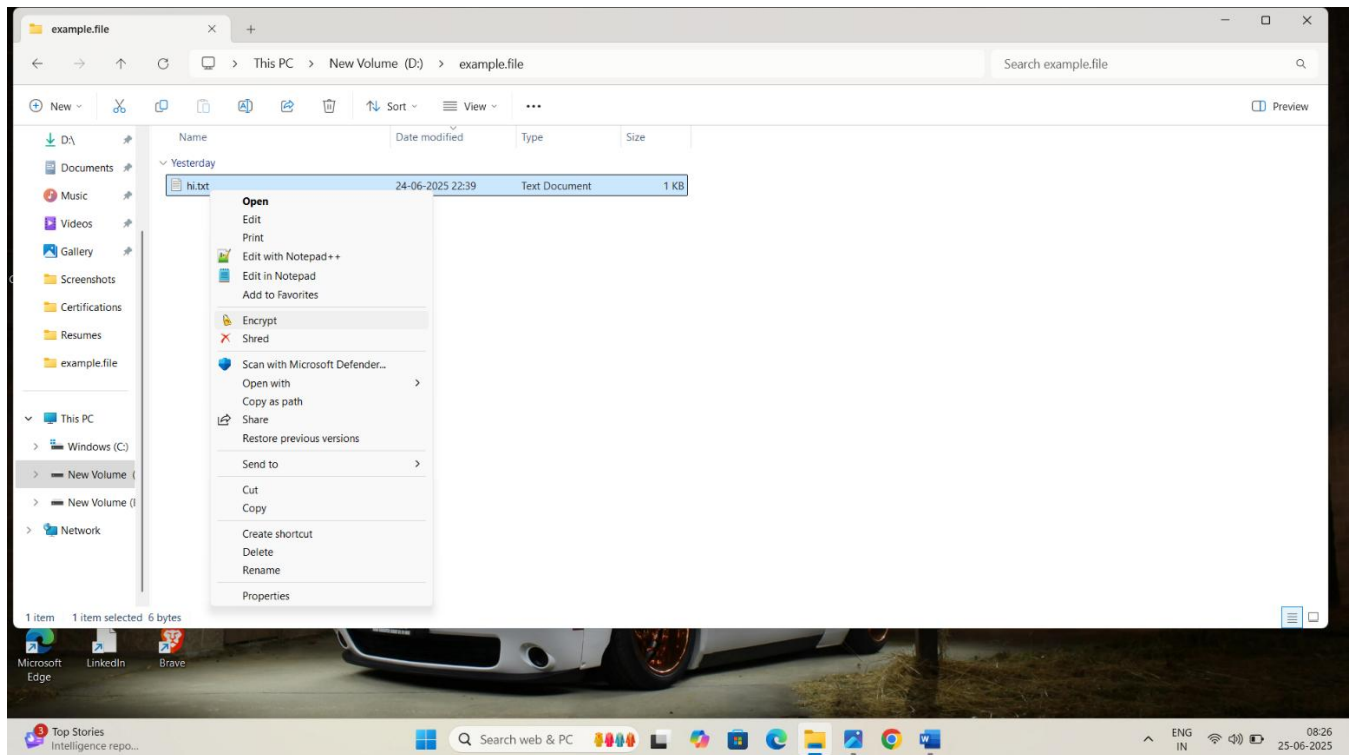


4th step : click on file section and select file which you want to encrypt

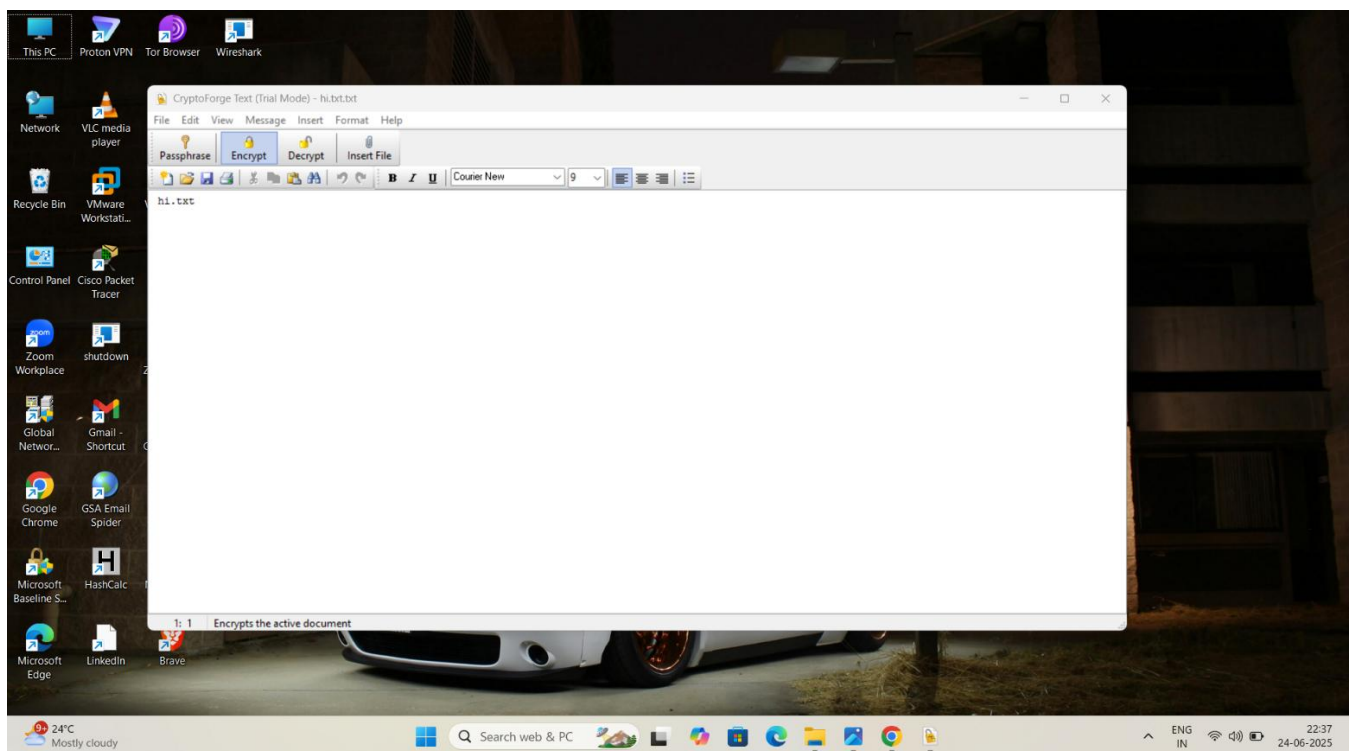


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5th step : right click on the file and select encrypt



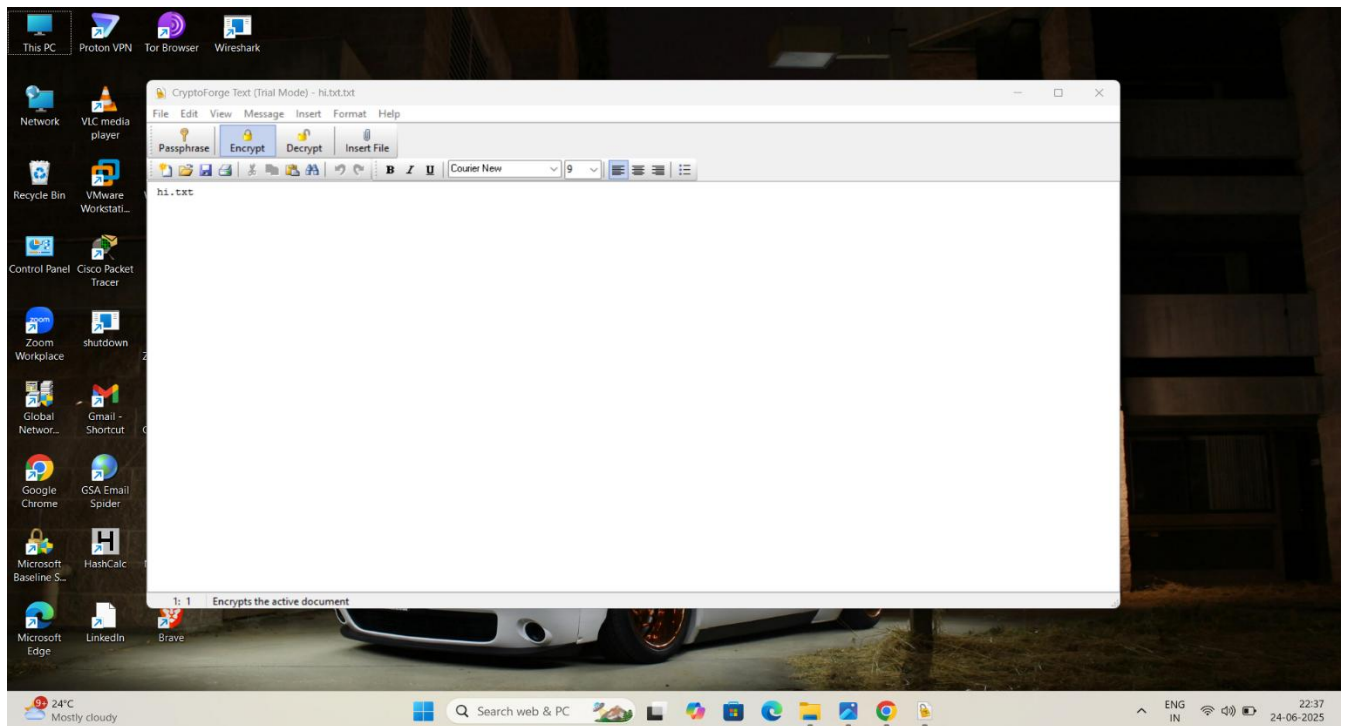
6th step : you can see the txt file on page now like this :



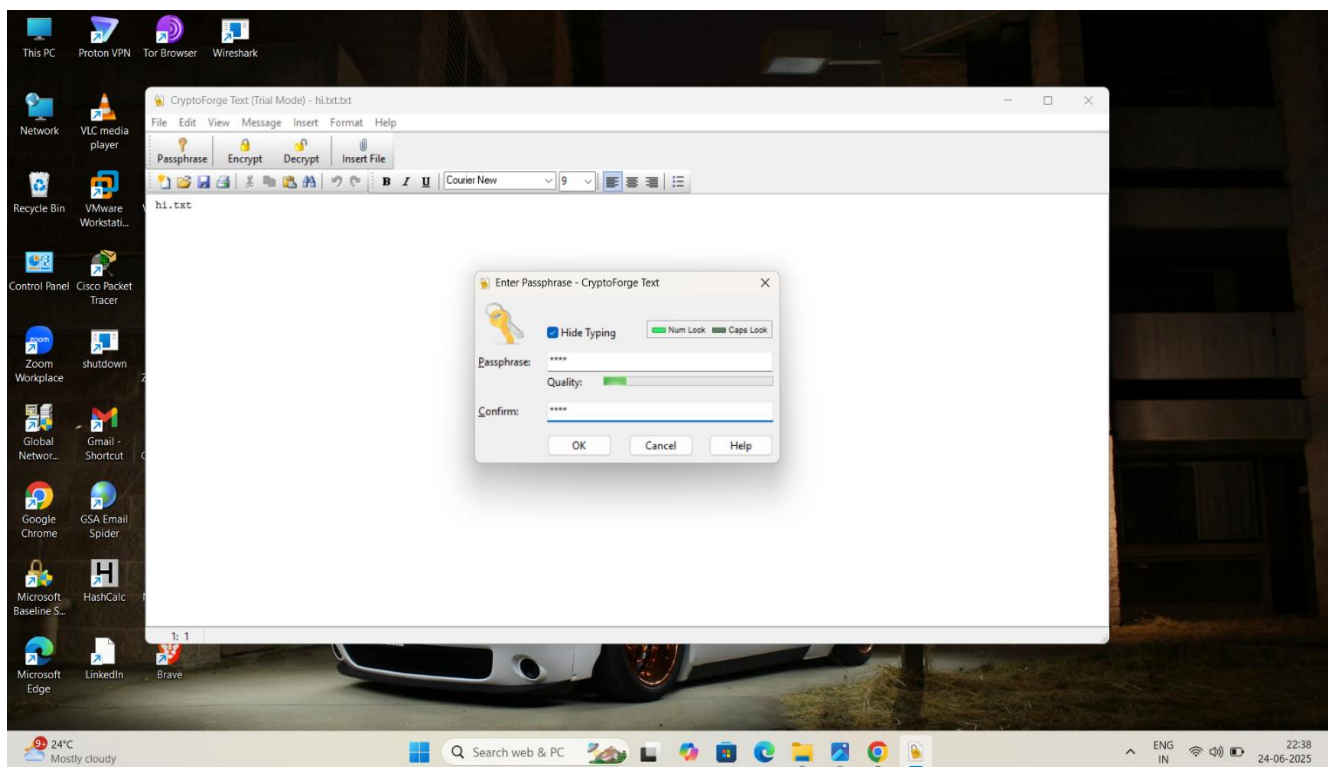
This text is inside the file.

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7th step : click on encrypt

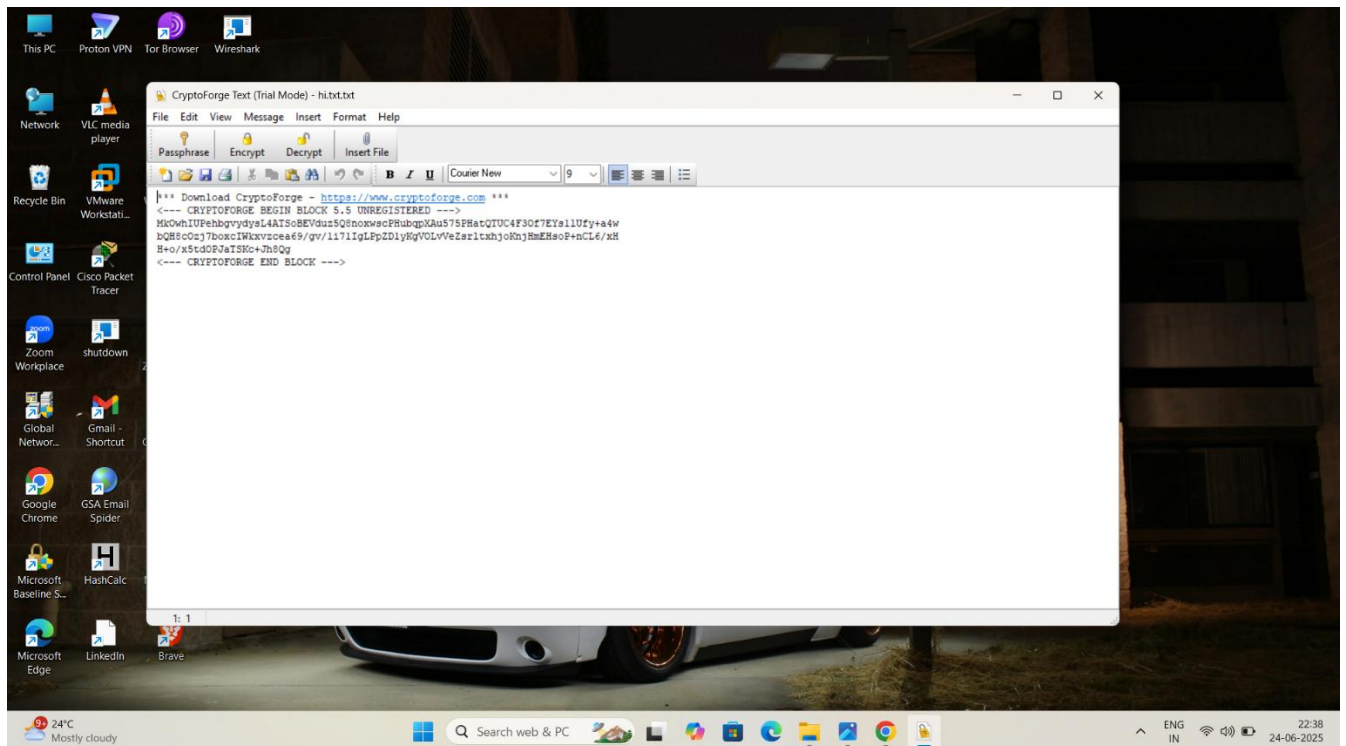


8th step : after you click on encrypt now you can set the password for encryption :

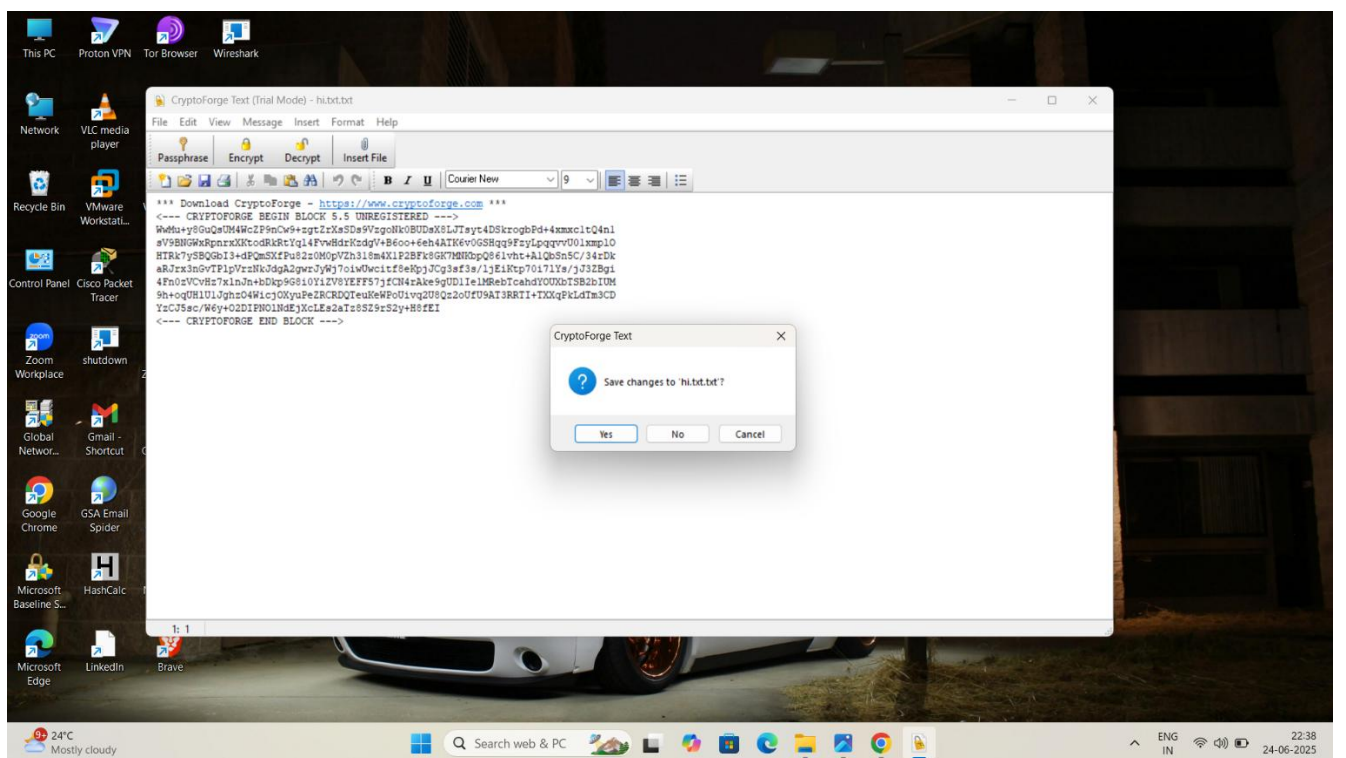


And click on ok

9th step : after you click on yes msg encrypt and see like this :

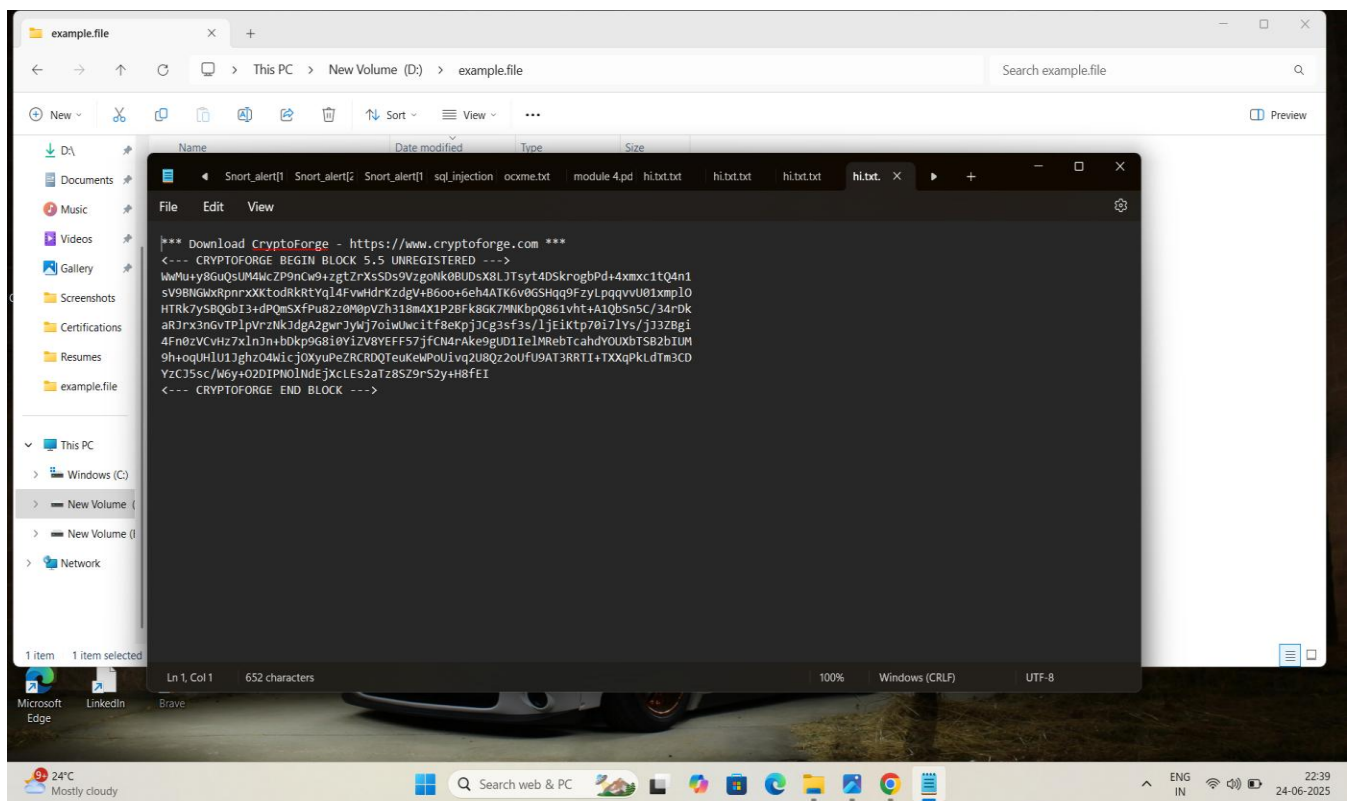


10th step : go back and click on yes for save changes :



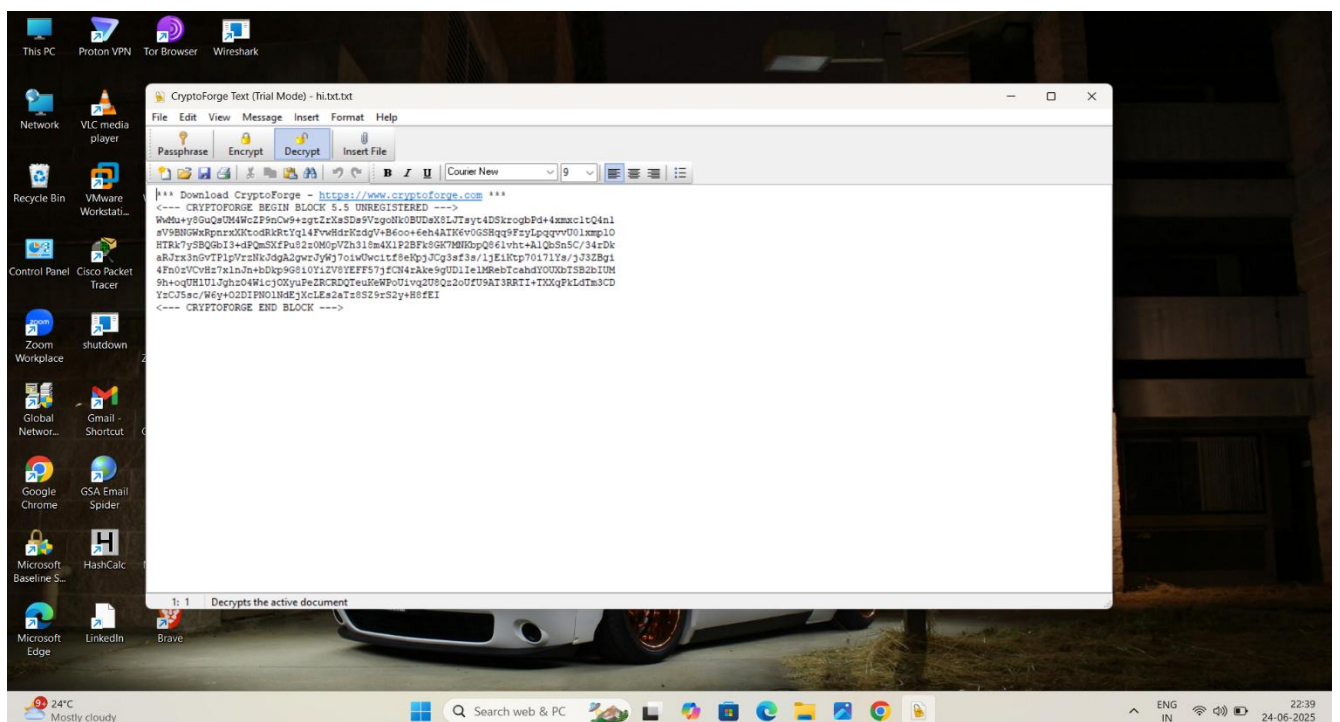
After this if you want to see the file : file is encrypted

11th step : go again this file and check file is encrypted or not



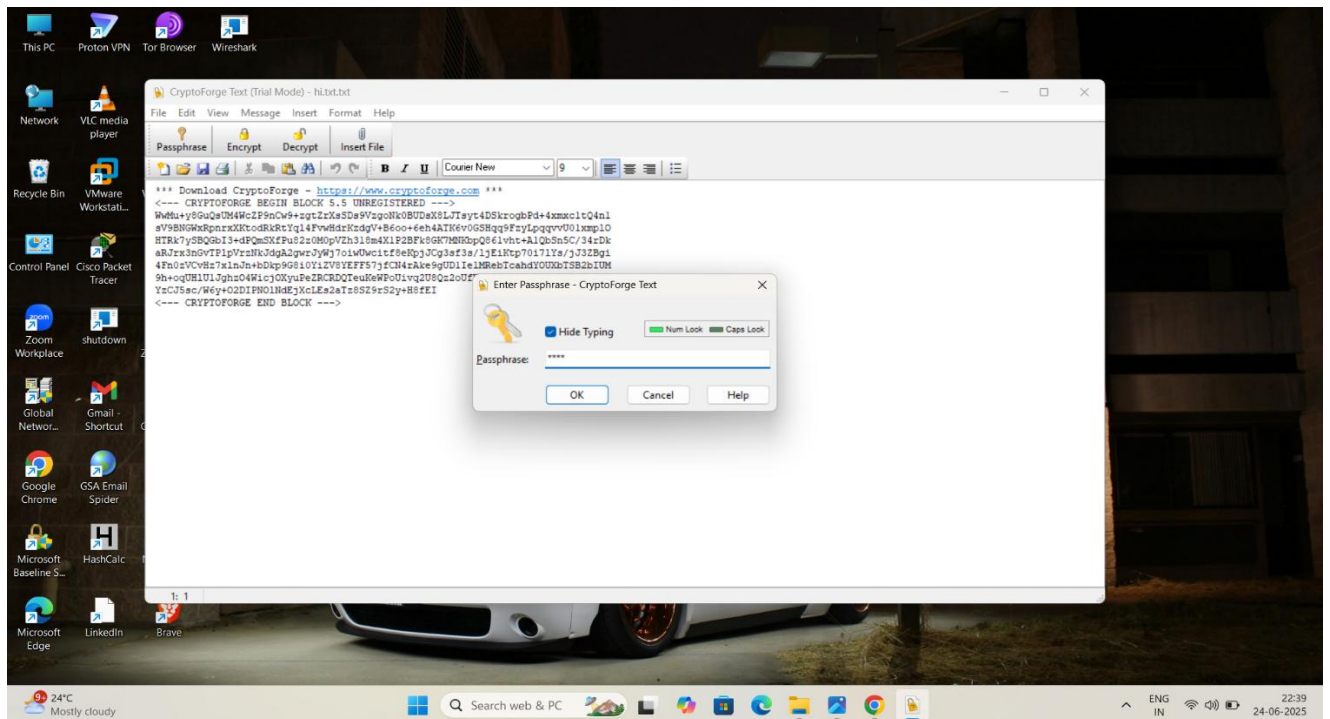
See file is encrypted

12th step : if you want to decrypt the file you can again right click on file and select encrypt then you come to cryptoforge interface

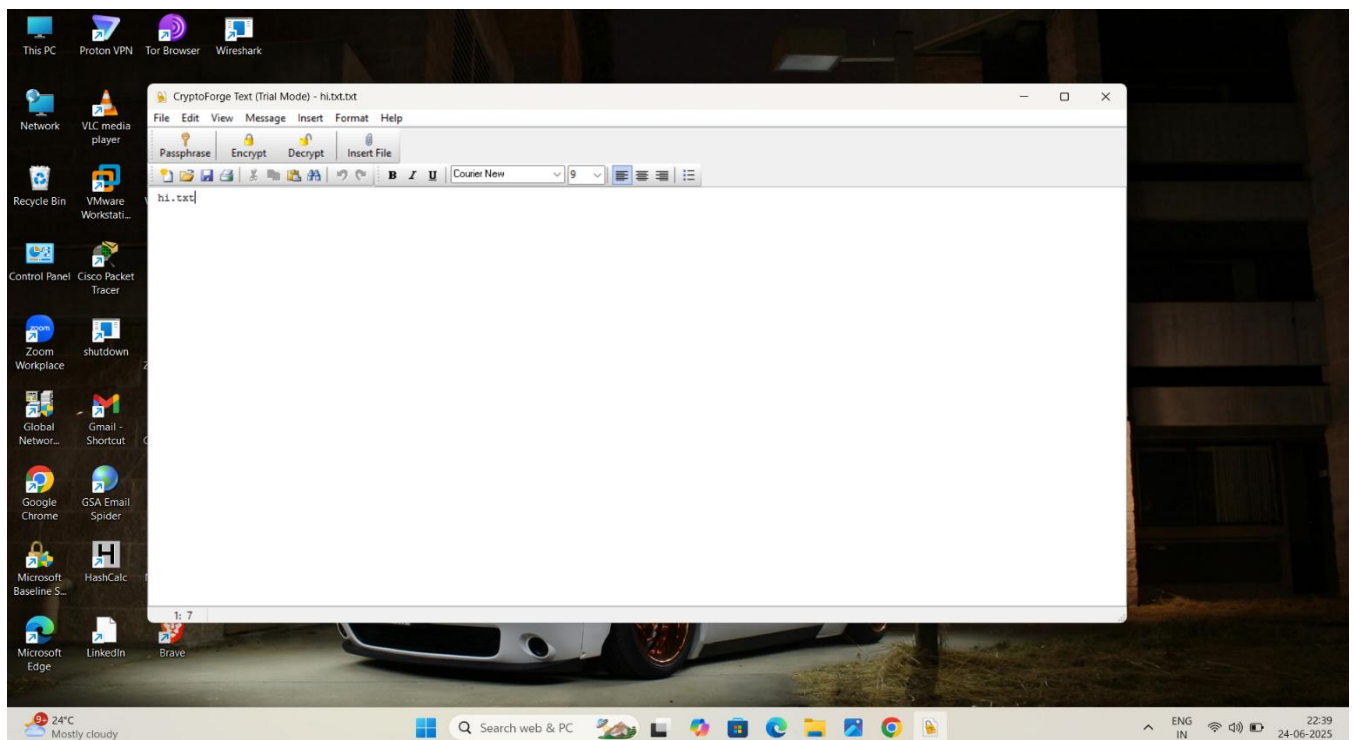


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13th step : click on decrypt



14th step : write the password for we use in for encrypt the file after then file is decrypt



See the file is decrypt successfully.

❖ Perform disk encryption using veracrypt

VeraCrypt is a free, open-source disk encryption software used to secure sensitive data by encrypting entire drives, partitions, or creating virtual encrypted volumes. It's a successor to the now-discontinued **TrueCrypt**, and is widely used for personal and professional data protection.

Key Features of VeraCrypt

- 1. Strong Encryption Algorithms**
 - AES (Advanced Encryption Standard)
 - Serpent
 - Twofish
 - Combination modes (e.g., AES-Twofish-Serpent)
 - 2. Types of Encryption**
 - **File container:** Acts like a virtual encrypted disk.
 - **Partition encryption:** Encrypts a specific disk partition.
 - **Full disk encryption (FDE):** Encrypts the entire operating system drive, including the boot partition.
 - 3. Hidden Volumes and OS**
 - Supports creation of **hidden volumes** and **hidden operating systems** for plausible deniability in high-risk environments.
 - 4. Cross-Platform Support**
 - Available on **Windows**, **Linux**, and **macOS**.
 - 5. On-the-fly Encryption**
 - Data is automatically encrypted and decrypted as it's accessed—transparent to the user.
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How VeraCrypt Works (Basic Steps)

- 1. Download and Install**
 - Available at: <https://www.veracrypt.fr>
- 2. Create a Volume**
 - Choose between:

- Encrypted file container
- Encrypt a non-system partition/drive
- Encrypt the system drive (with pre-boot authentication)

3. Volume Creation Wizard

- Select volume type (Standard or Hidden)
- Set volume location
- Choose encryption algorithm
- Define volume size and password
- Format the volume (FAT/exFAT/NTFS)

4. Mount the Volume

- Use VeraCrypt to mount the volume
- Enter password and access the encrypted files like a normal drive

5. Dismount When Done

- Dismount the volume to secure your data

Advantages

- Free and open-source
- Very secure (multiple algorithms)
- Supports plausible deniability
- Portable mode (can run without installing)

Limitations

- No recovery if password is lost
- May be slower than unencrypted drives (slightly)
- Not suitable for cloud sharing directly (encrypted containers must be uploaded/downloaded)

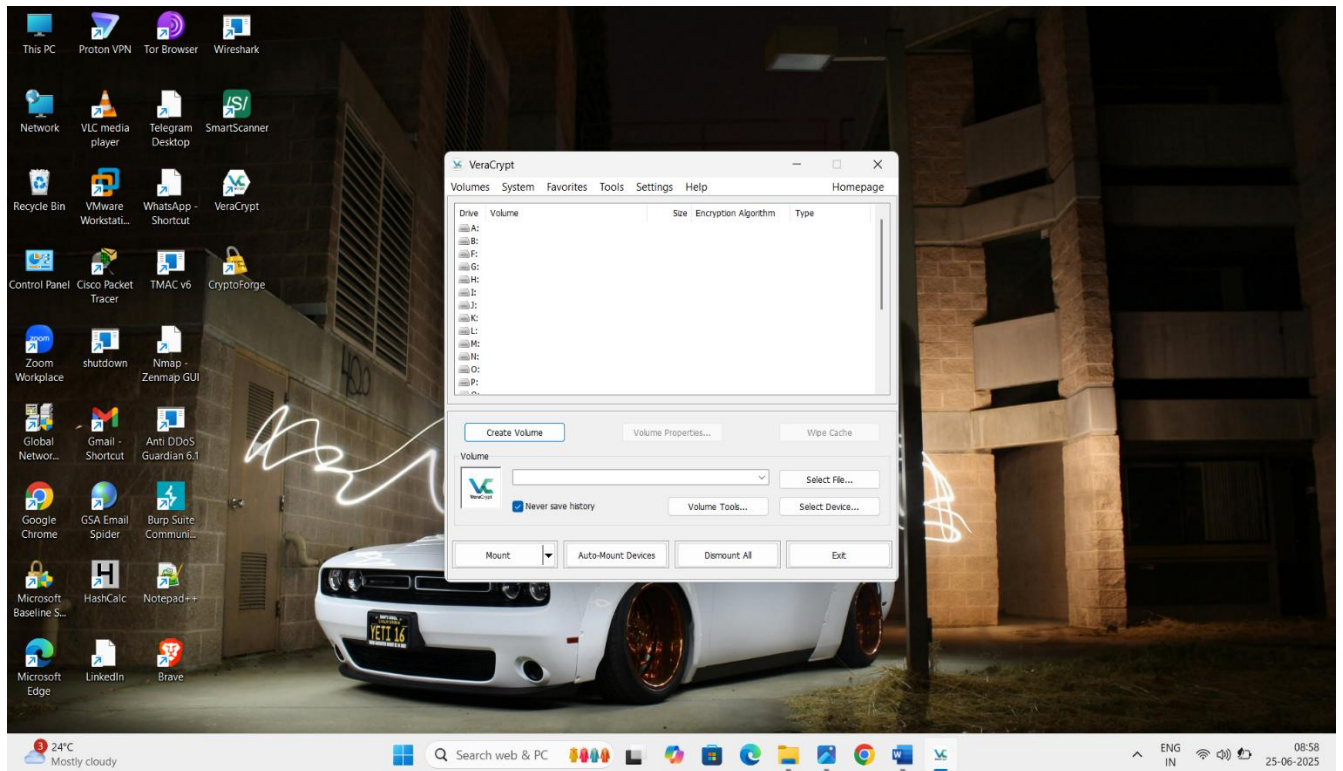
Use Case Examples

- Protecting personal data (bank details, ID scans)
- Securing external USB drives
- Full disk encryption for laptops
- Creating hidden storage for sensitive files

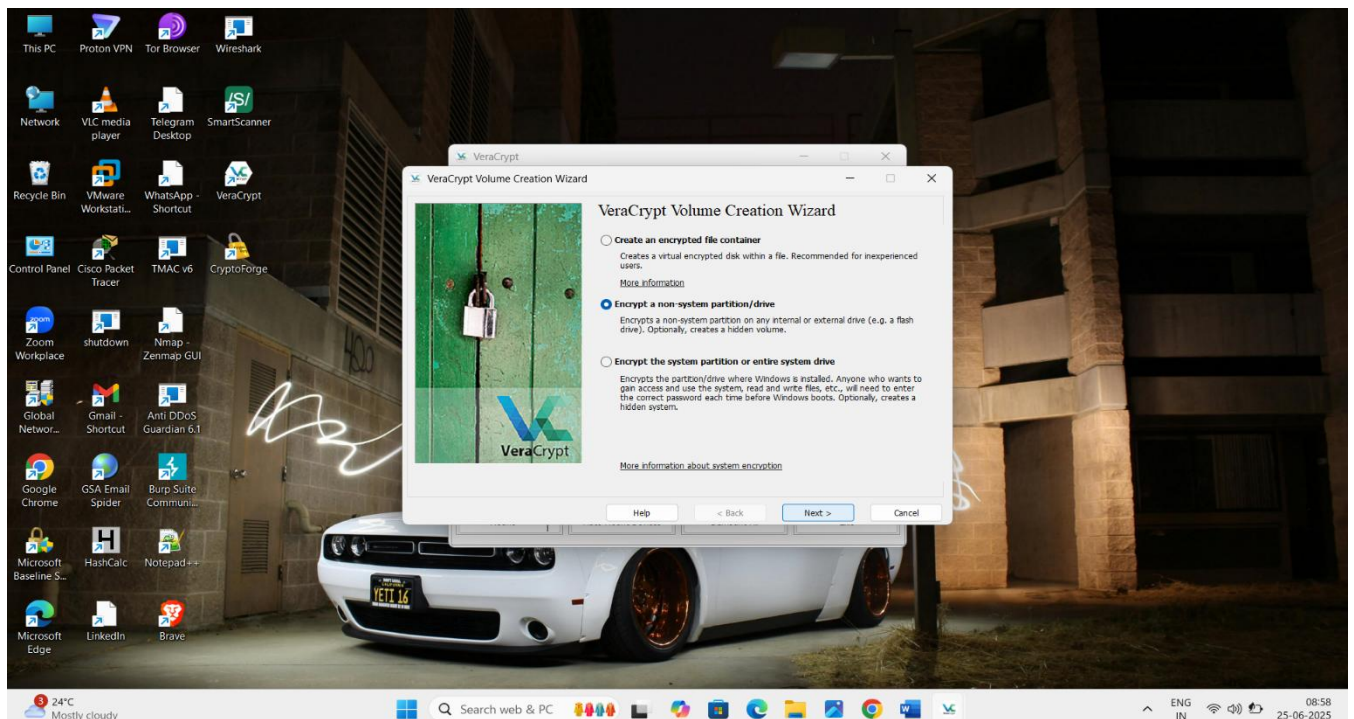
How to use :

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1st step : install and open veracrypt

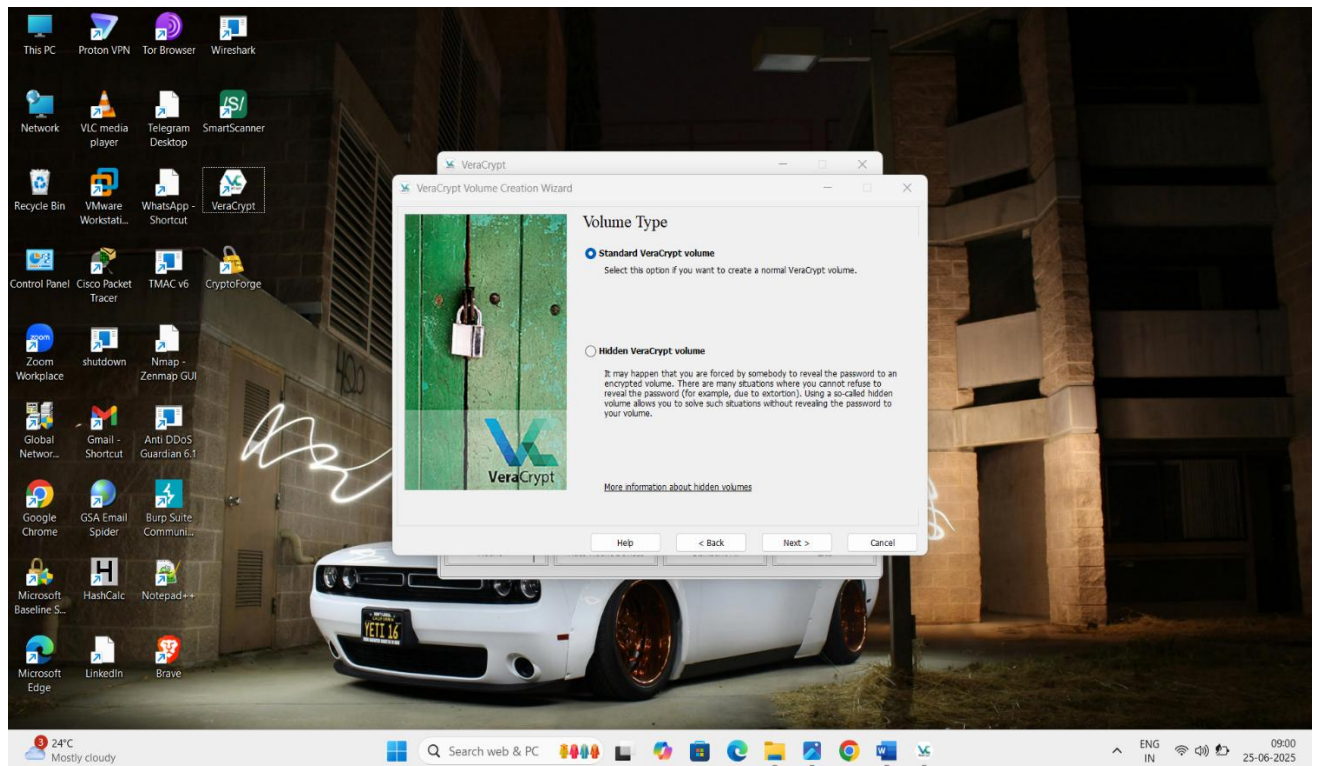


2nd step : click on create volume

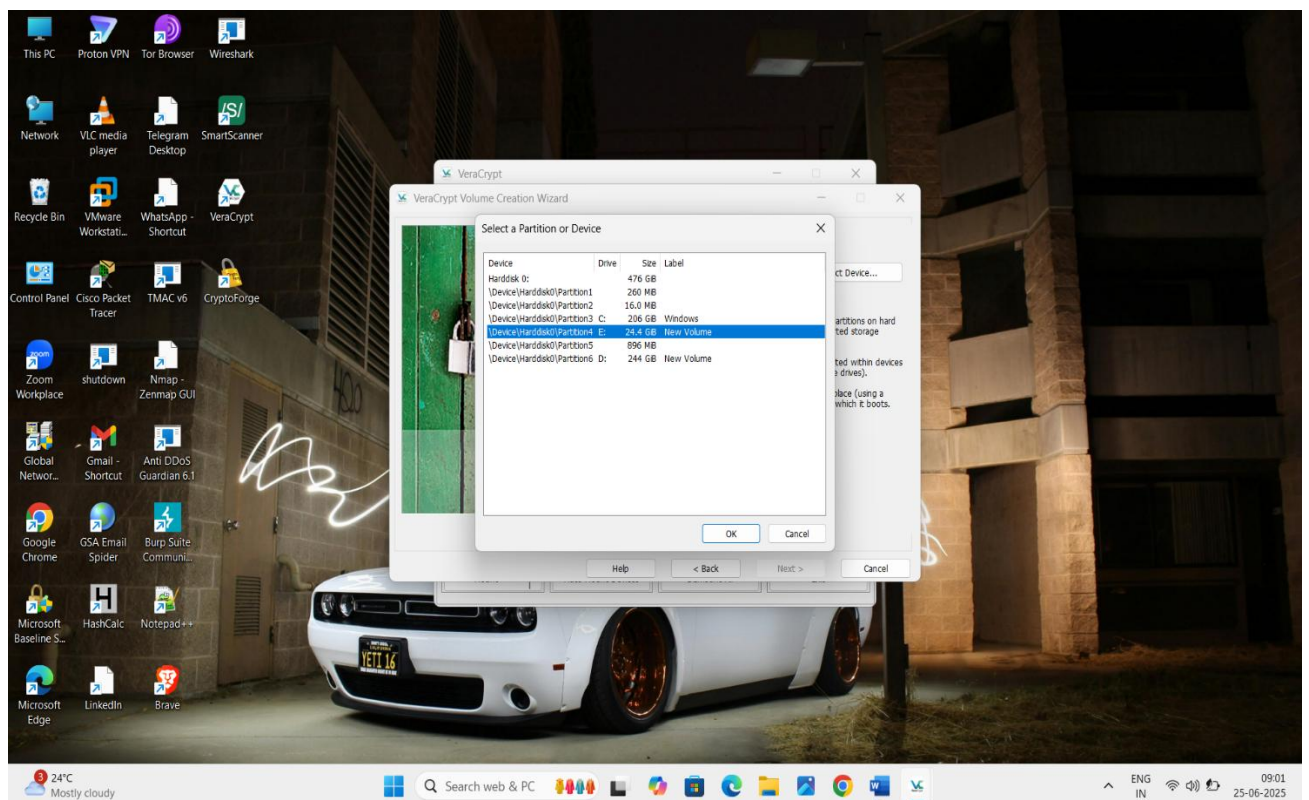


You can see the 3 option here

3rd step : select 2nd option and click on next

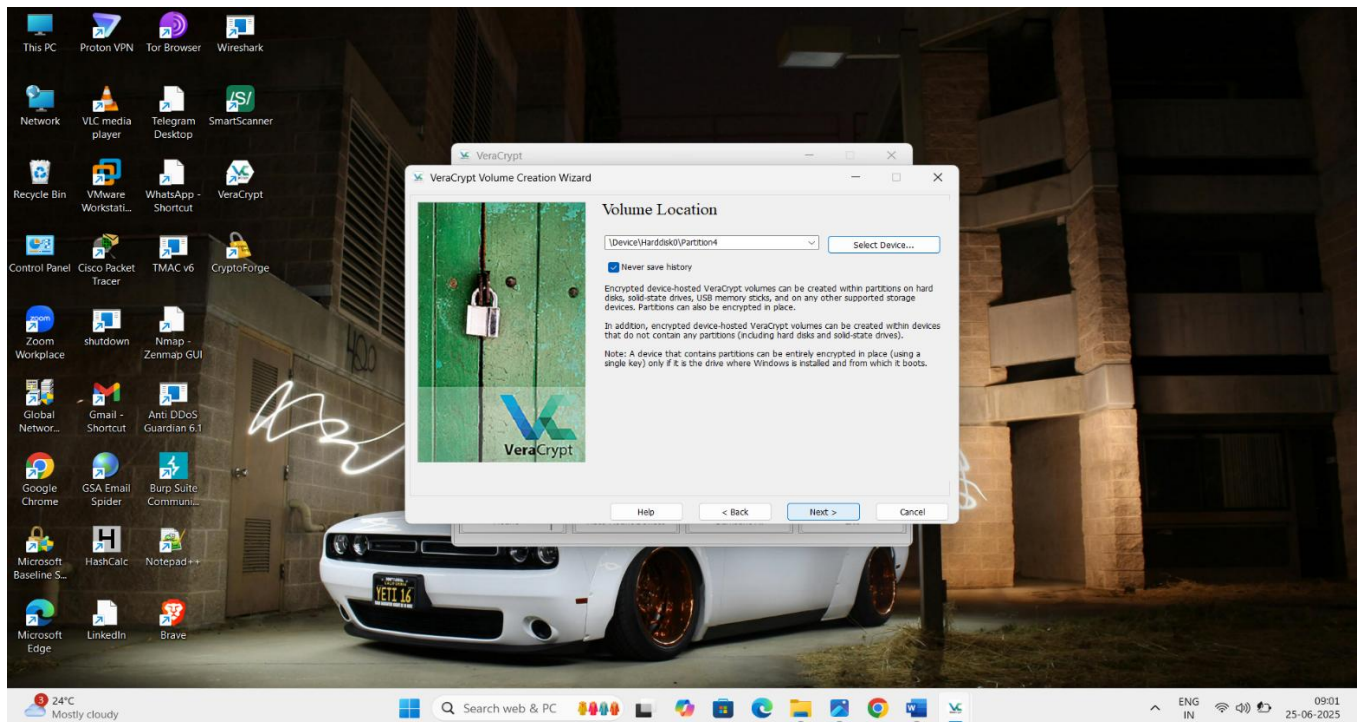


4th step : select first option and click on next

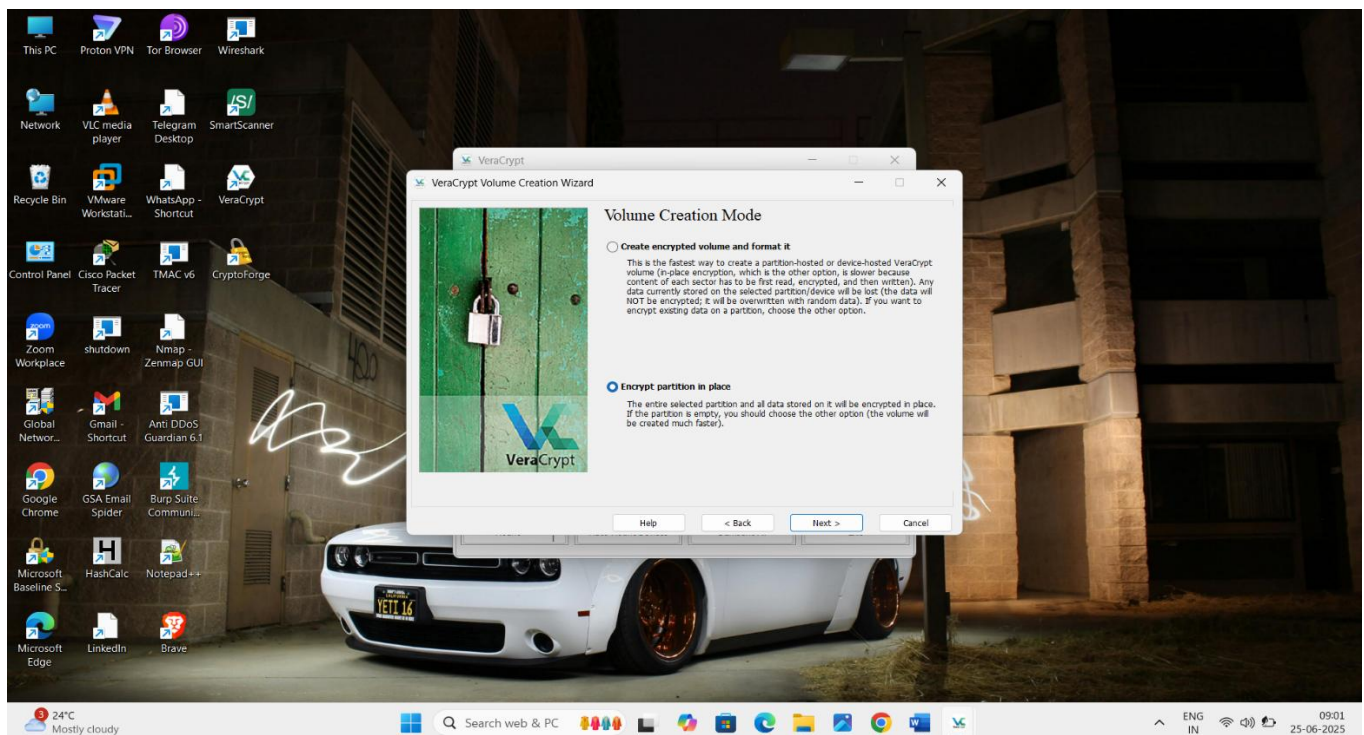


Select which drive you need to hide or extract

5th step : click on ok



6th step : click on next



Select 2nd option and click on next the drive is incrypted successfully now .

❖ Perform file Encryption using private Encrypter

🔑 What is a Private Encrypter?

A **private encrypter** usually refers to an **encryption tool or algorithm** that uses a **private (secret) key** to encrypt and decrypt data. This term is often associated with **symmetric encryption**, where the **same key is used** for both encryption and decryption.

💡 Types of Encryption (Context)

To understand "private encrypter" better, you should know the two main types of encryption:

Type	Key Used	Description
Symmetric Encryption	One private key	Same key encrypts and decrypts the message. Fast and efficient.
Asymmetric Encryption	Public and private keys	Public key encrypts, private key decrypts. More secure for data sharing.

🔧 How a Private Encrypter Works (Symmetric Example)

1. **Key Generation:** A secret key is generated.
2. **Encryption:**
 - Plaintext (original message) + Key → Ciphertext (encrypted message)
3. **Decryption:**
 - Ciphertext + Same Key → Plaintext

🔑 If someone gets the private key, they can decrypt all data. That's why it must be kept secret.

🔑 Common Symmetric Algorithms (used in private encryption)

- **AES (Advanced Encryption Standard)**

- **DES (Data Encryption Standard)**
 - **Blowfish**
 - **ChaCha20**
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Real-Life Example

Imagine a zip file encrypted with a password. Only people with the password (private key) can open it. That's a simple form of private encryption.

In Cybersecurity or Ethical Hacking

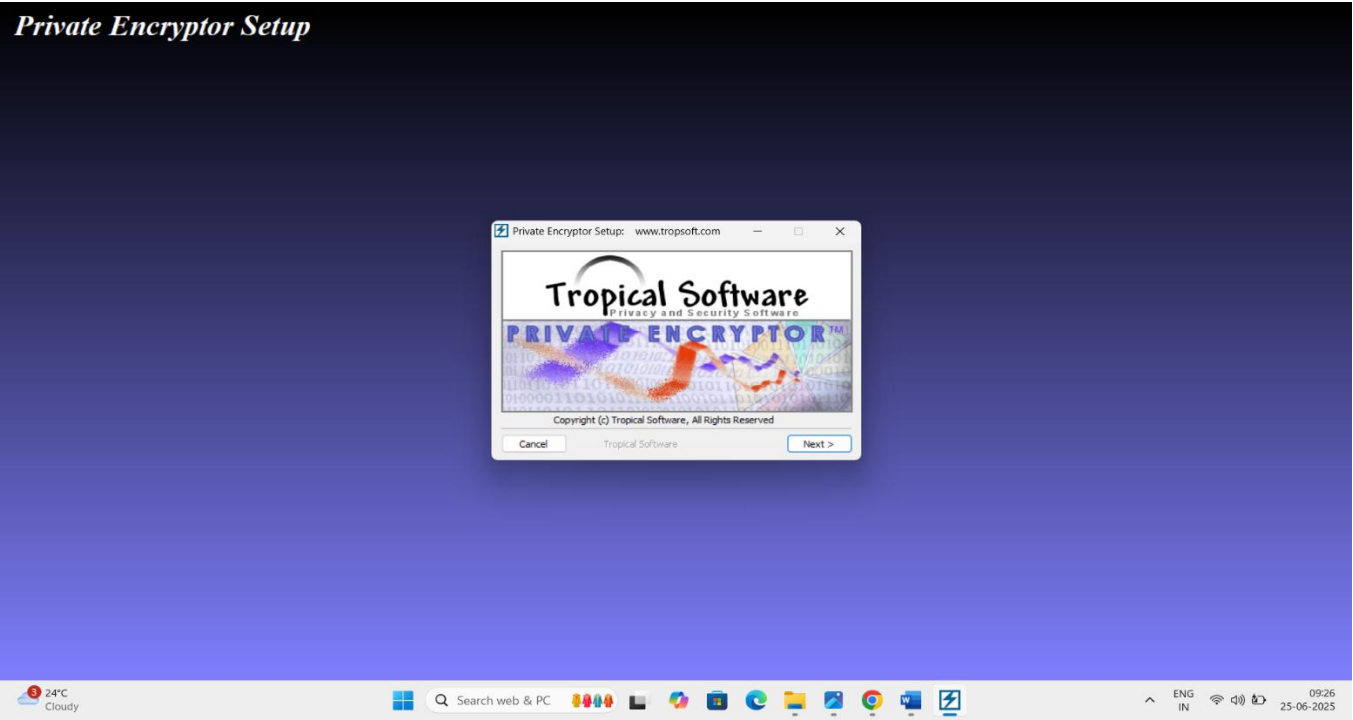
A **private encrypter** might be a **custom tool** used by hackers or security professionals to:

- Encrypt payloads (to bypass antivirus)
- Protect scripts from reverse engineering
- Secure communication in malware or pentesting tools

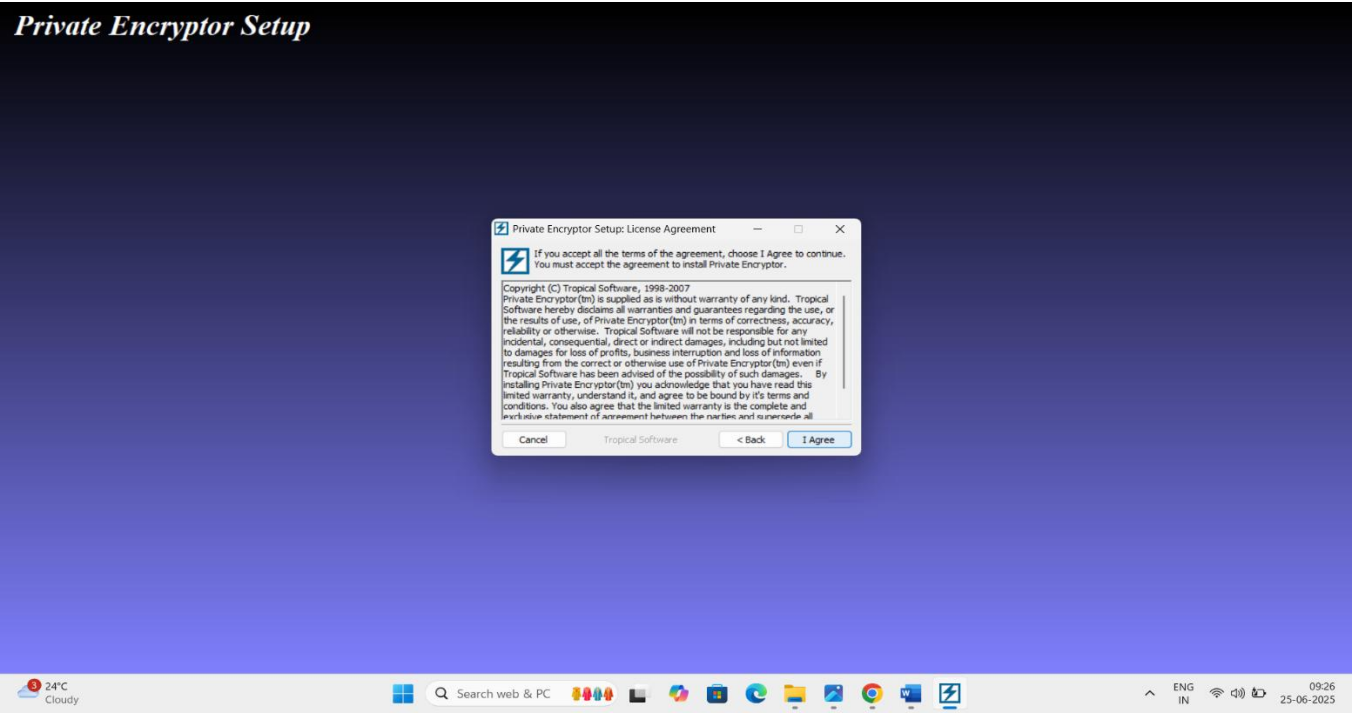
 **Note:** In malicious use, attackers may create "private encrypters" to hide malware or ransomware code from detection.

How to use for Encryption :

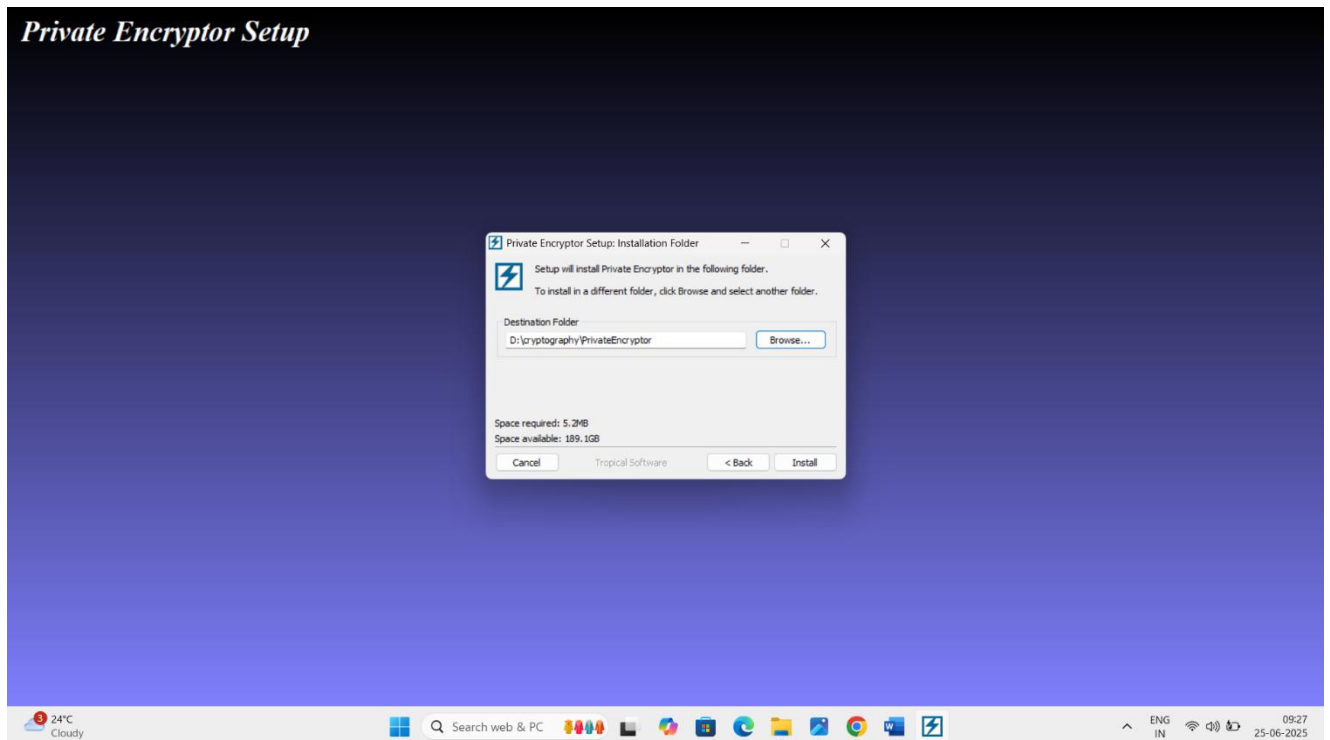
1st step : install private encrypter and run



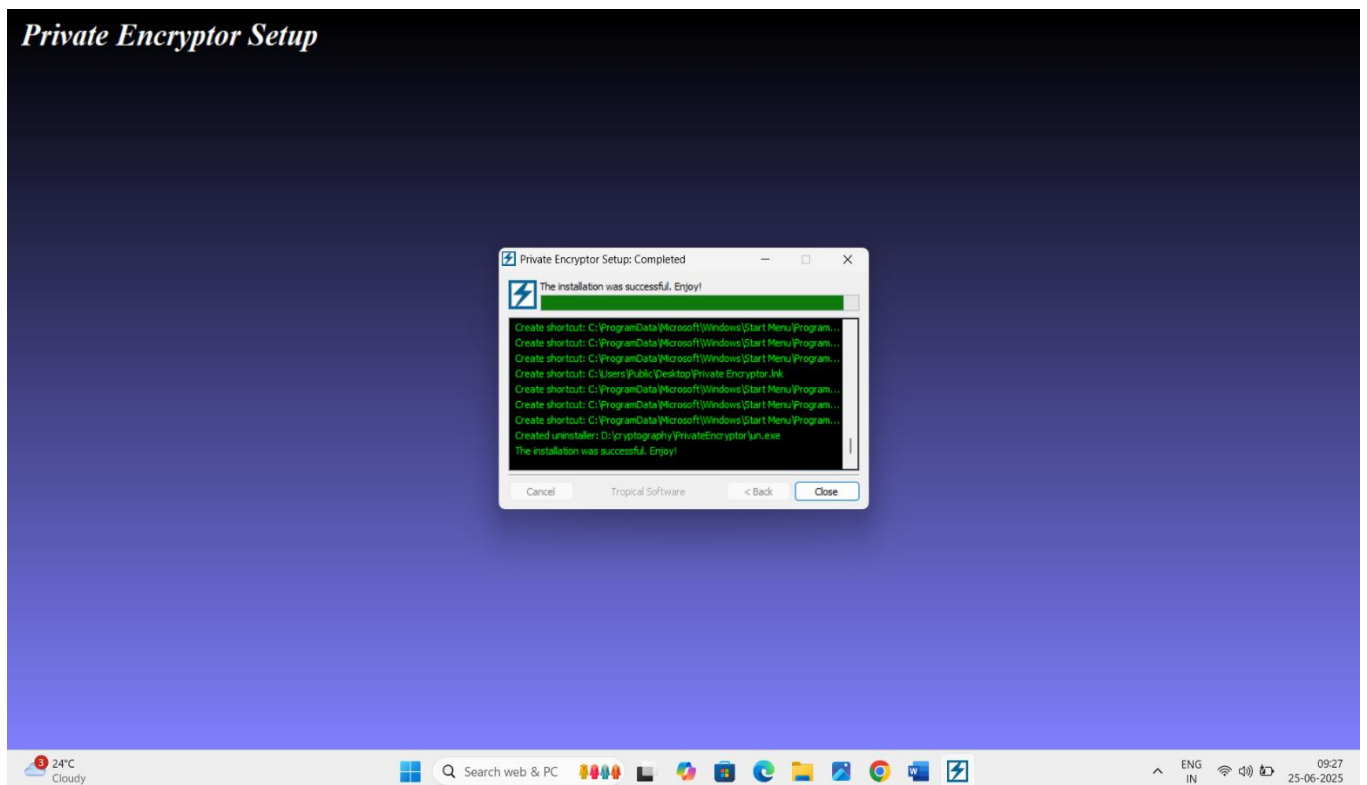
Click on next



Click on I Agree

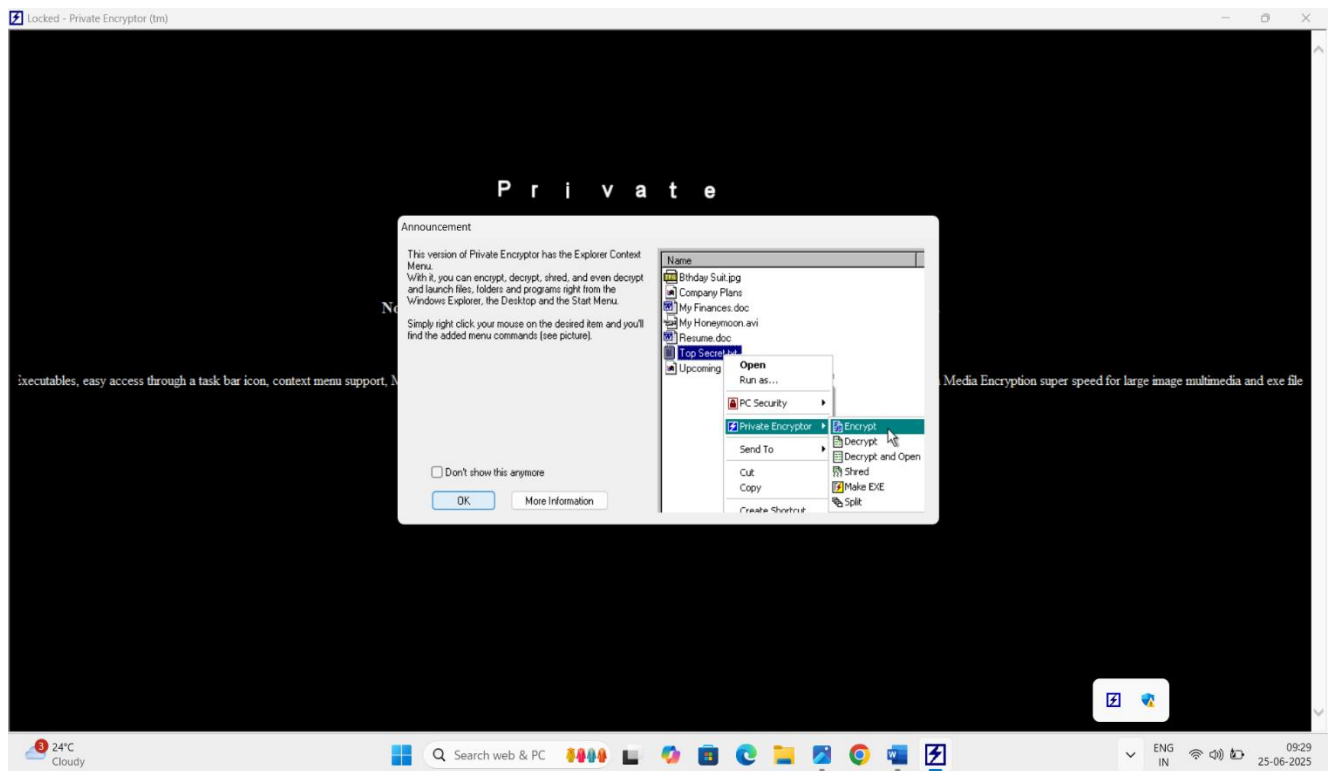


Give the file location which you want and click on Install



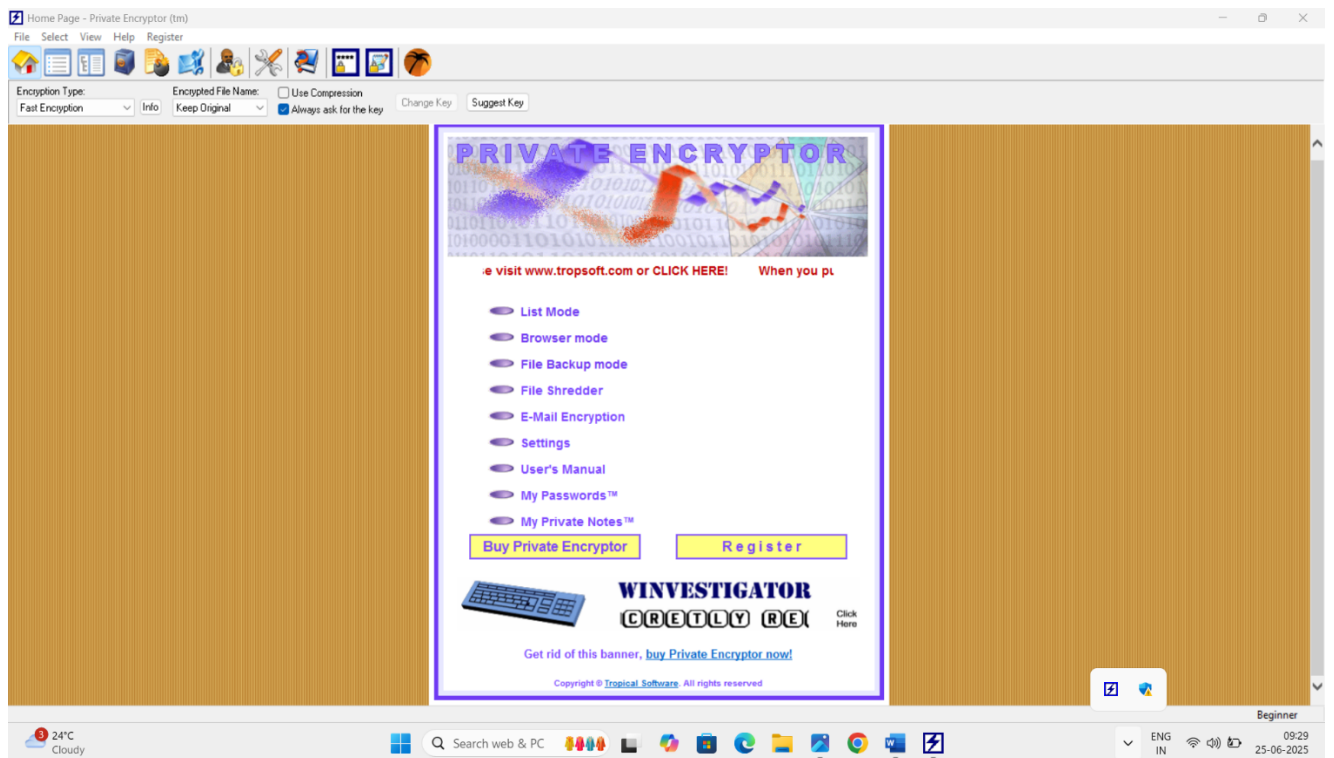
You can see the installation is done now close the tab

Double tab on private encrypter to open



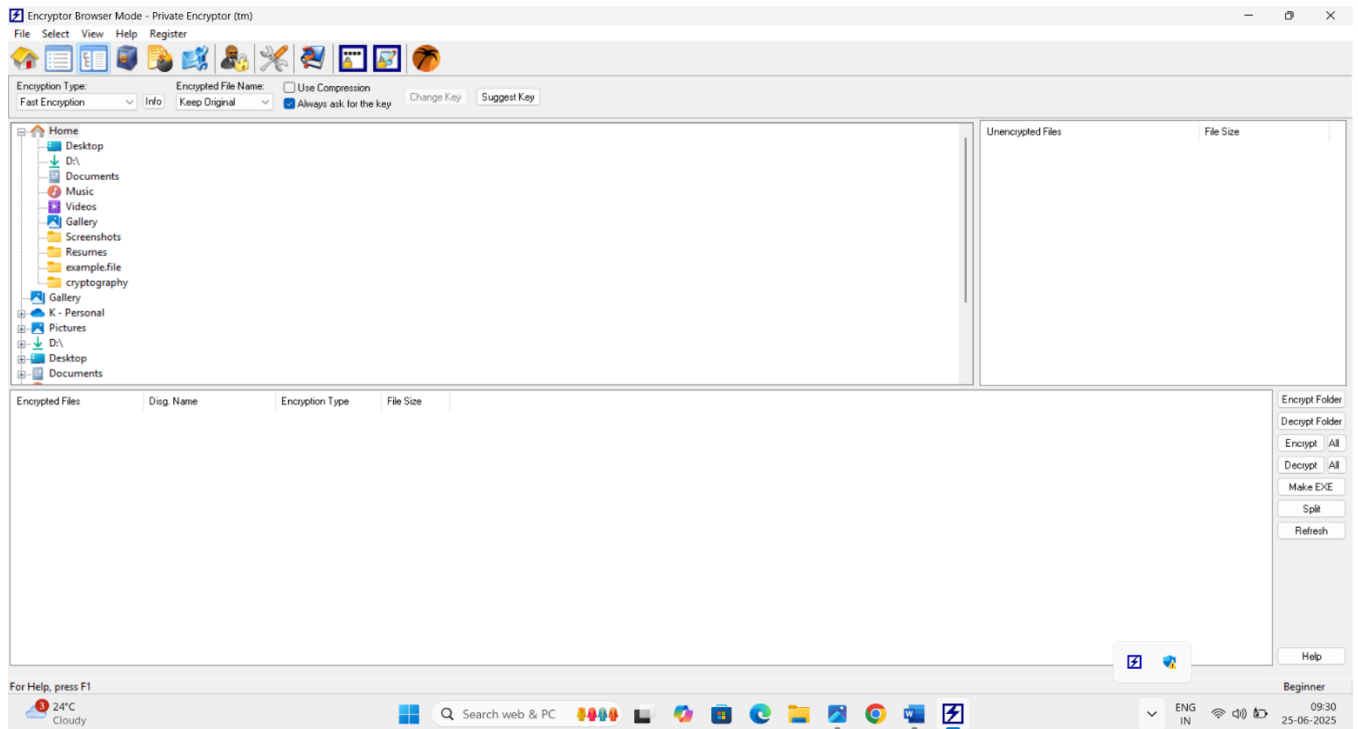
You can see this page tutorial

Click on OK

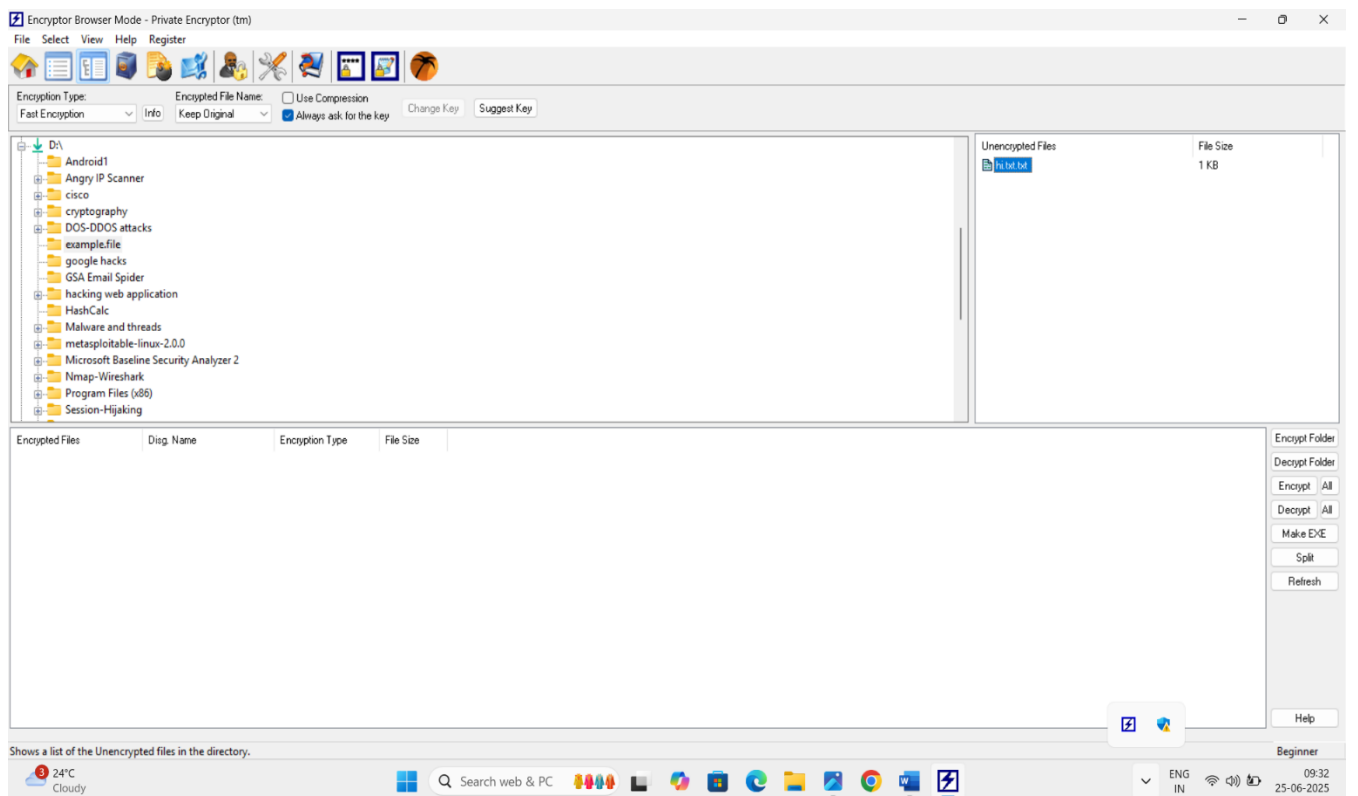


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Click on list mode



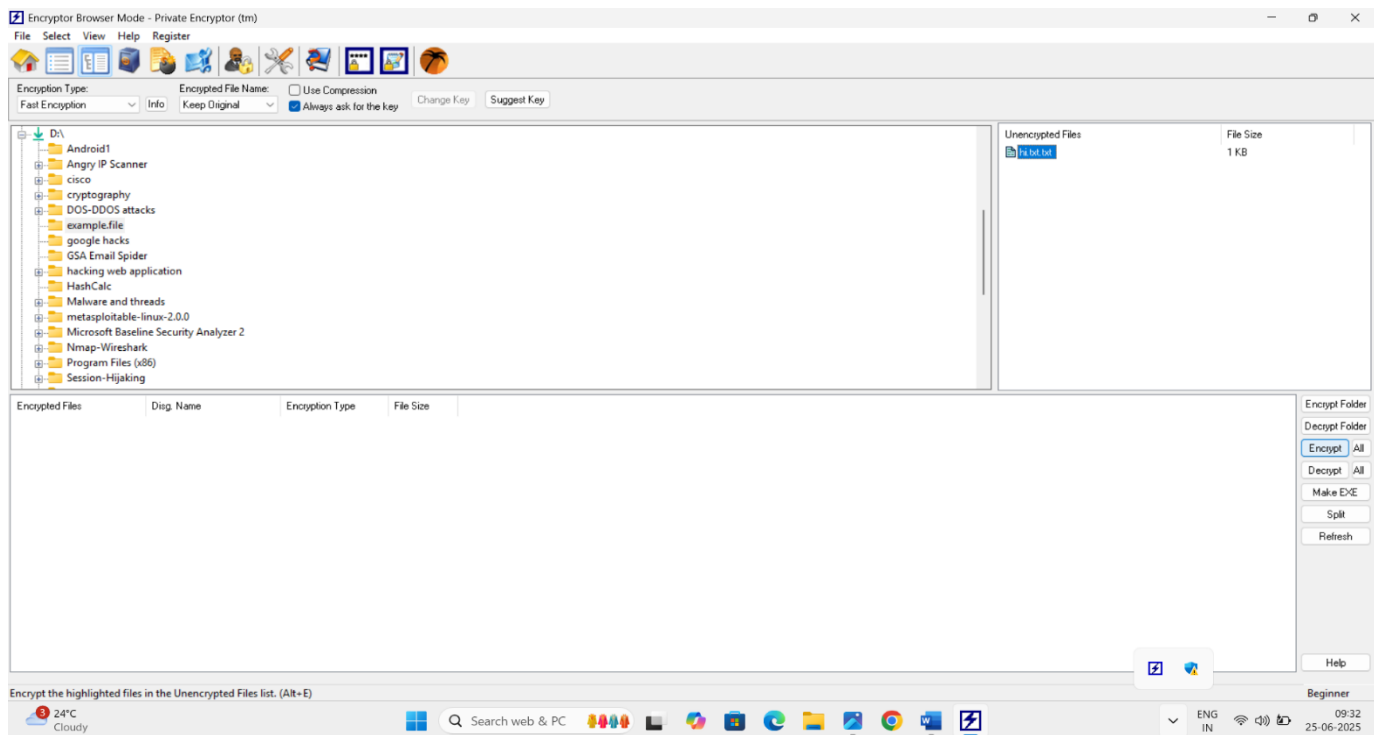
Select the file which we want to Encrypt



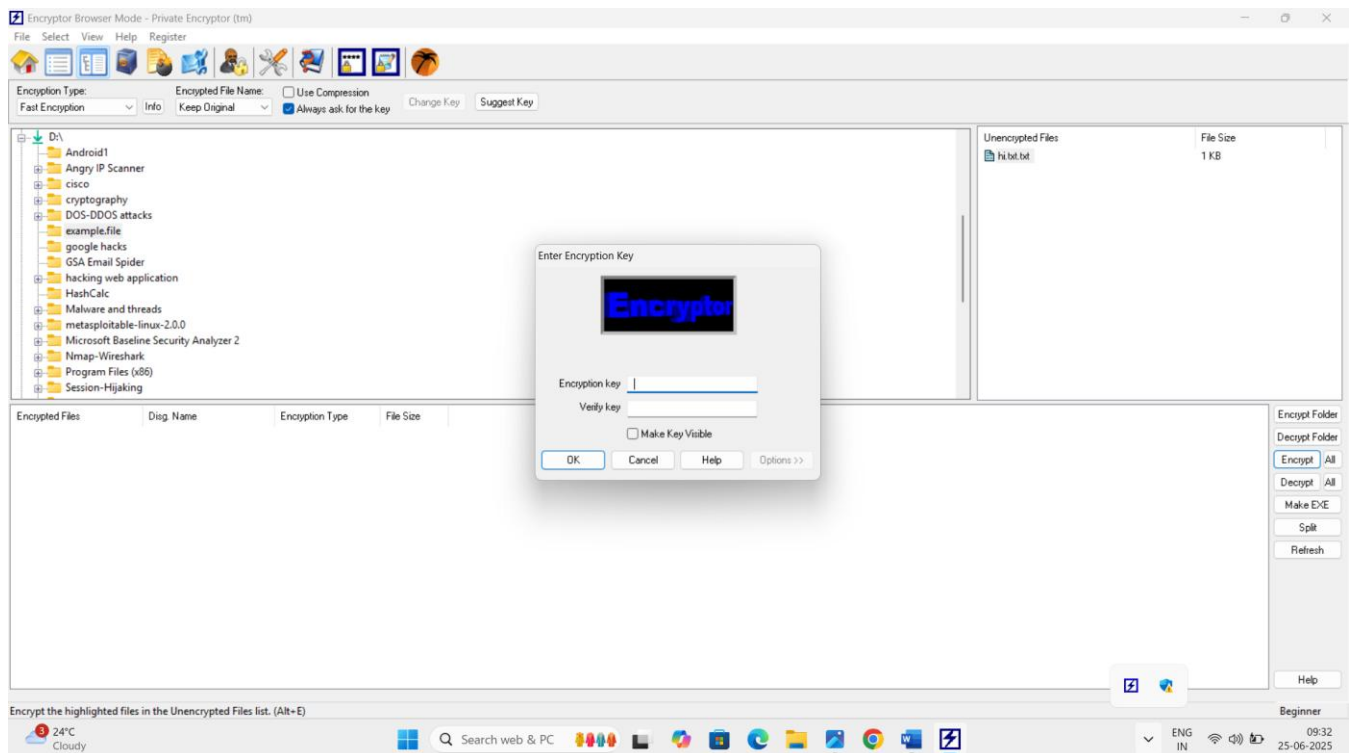
On the right side you see the selected file

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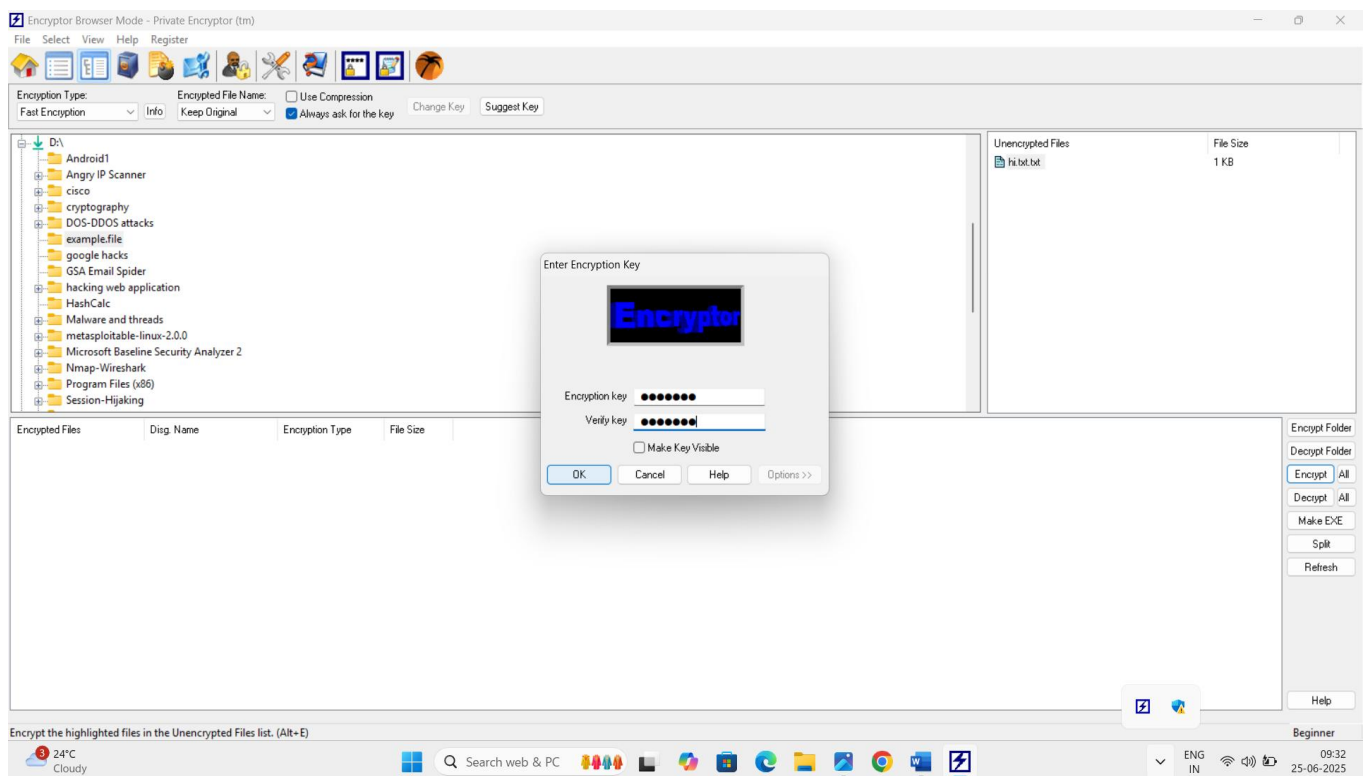
Select the file and click on Encrypt for Encrypting



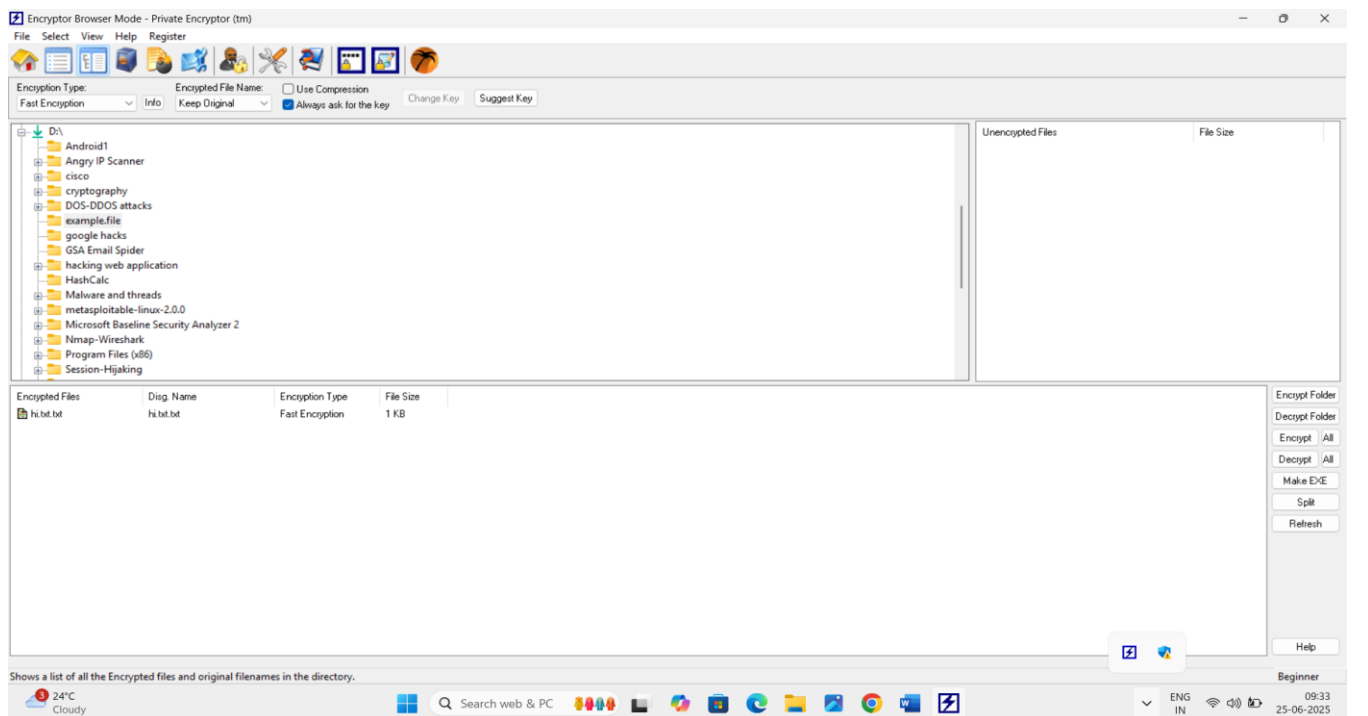
After this on screen come a password popup like this :



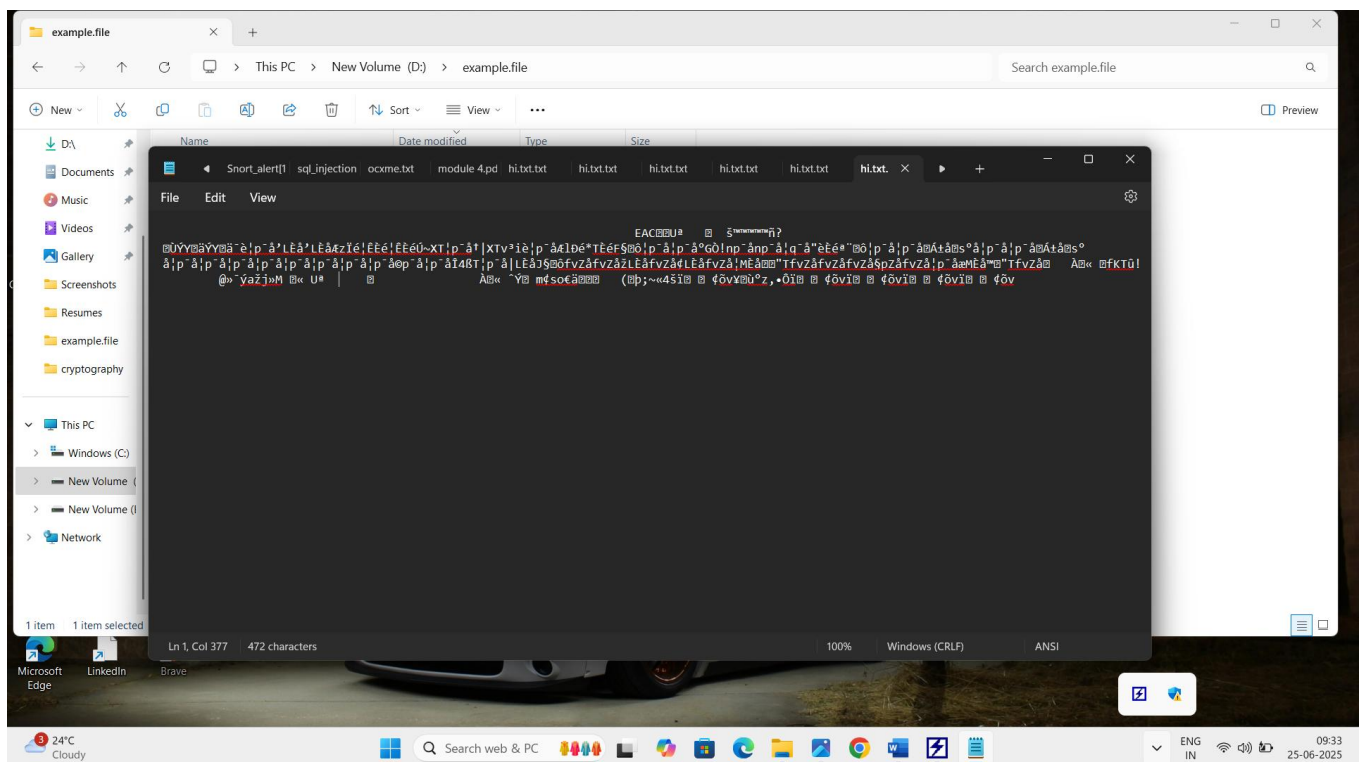
Write the password which you set for a password Encrypter and alick on ok



You can see in downside of the interface file encrypted

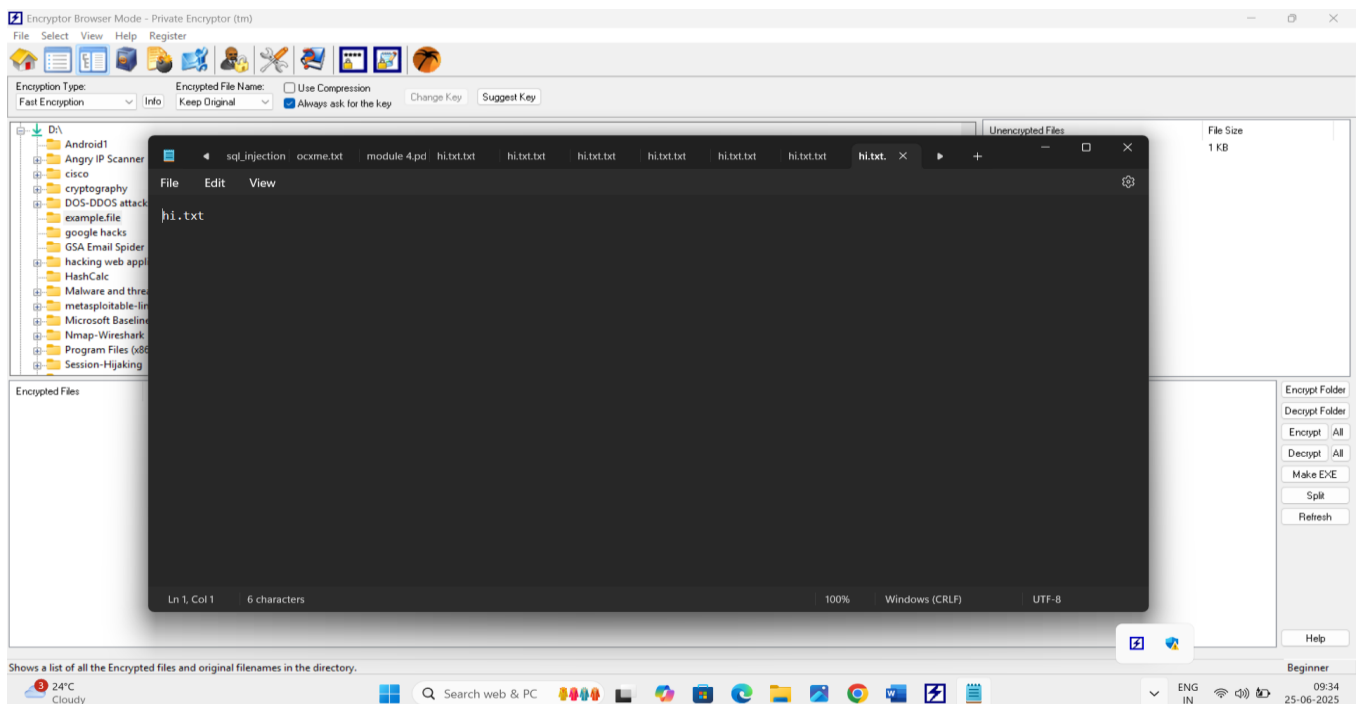


To see the file was encrypt or not go to file destination and open the file which you encrypted



See the file is in hash format it means the file is encrypt successfully.

If you want to decrypt the file go in private encrypter interface and select the file and click on decrypt



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See the file in clear format that means file decrypted .

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